HEART RATE CHANGES AS AN INDICATOR OF ACADEMIC STRESS IN COLLEGE STUDENTS

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By

TANJIDUL HUDA

Under the supervision of
Dr.Neha Singh
Dr.Aparna Dixit

Department of Biomedical Science
Bhaskaracharya College of Applied Sciences

University of Delhi

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Certificate

This is to certify that the Dissertation entitled “HEART RATE CHANGES AS AN INDICATOR OF ACADEMIC STRESS IN COLLEGE STUDENTS ” submitted by ‘TANJIDUL HUDA’ in partial fulfilment of the requirements for the degree of B. Sc. (Hons) Biomedical Science with Research is a Bonafide work carried out under my/our supervision.

Date:

Dr.Neha Singh
Dr.Aparna Dixit

Declaration

I hereby declare that this dissertation entitled “HEART RATE CHANGES AS AN INDICATOR OF ACADEMIC STRESS IN COLLEGE STUDENTS” is my original work and has been carried out under the guidance of “Dr.Neha Singh”, Name of the Department, Bhaskaracharya College of Applied Sciences, University of Delhi.

This work has not been submitted elsewhere for the award of any other degree or diploma.

Date:

TANJIDUL HUDA

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ABSTRACT

Stress is a key psychological and physiological problem that college students face, particularly in exams. Stress impacts cause measurable cardiovascular changes including the increase in heart rate and changes in heart rate variability (HRV). In the present study, heart rate variation has been employed to explore the physiological changes of the heart as an indicator of academic stress in college students by electrocardiography (ECG).

The study was carried out with healthy college students following the screening process with Perceived Stress Scale (PSS-10) questionnaire and health questionnaires. The ECG recordings were made under the normal resting condition with BIOPAC MP36 system and ECG-1 channel. Electrodes were positioned on standard anatomical sites such as right arm (RA), left arm (LA) and left leg (LL) using a modified Lead II configuration. Heart rate variability parameters were extracted from ECG signals with the software Kubios HRV Scientific Lite 4.3.0.

The parameters measured in the study were: mean heart rate, minimum heart rate, maximum heart rate, standard deviation, SDNN, RMSSD, LF, HF, and LF/HF ratio. ECG data was acquired and then processed and analyzed using Kubios software to assess the autonomic nervous system activity and cardiovascular variability.

The recordings taken on the normal days showed a normal level of cardiovascular activity with physiological changes in heart rate. There were technical problems with BIOPAC working during exam periods, so it was not possible to record during the examinations. The results of examinations are thus discussed as physiological reactions that are to be expected, given the literature results and theoretical knowledge of the physiology of stress.

It is expected that the heart rate, sympathetic activity, HRV, and autonomic balance will be different during examination stress conditions than resting conditions.

The study points to the potential of using ECG and HRV as objective and non-invasive tool to assess college students' academic stress.

Keywords: academic stress, ECG, BIOPAC, HRV, Kubios, Stress physiology.

Graphical Abstract

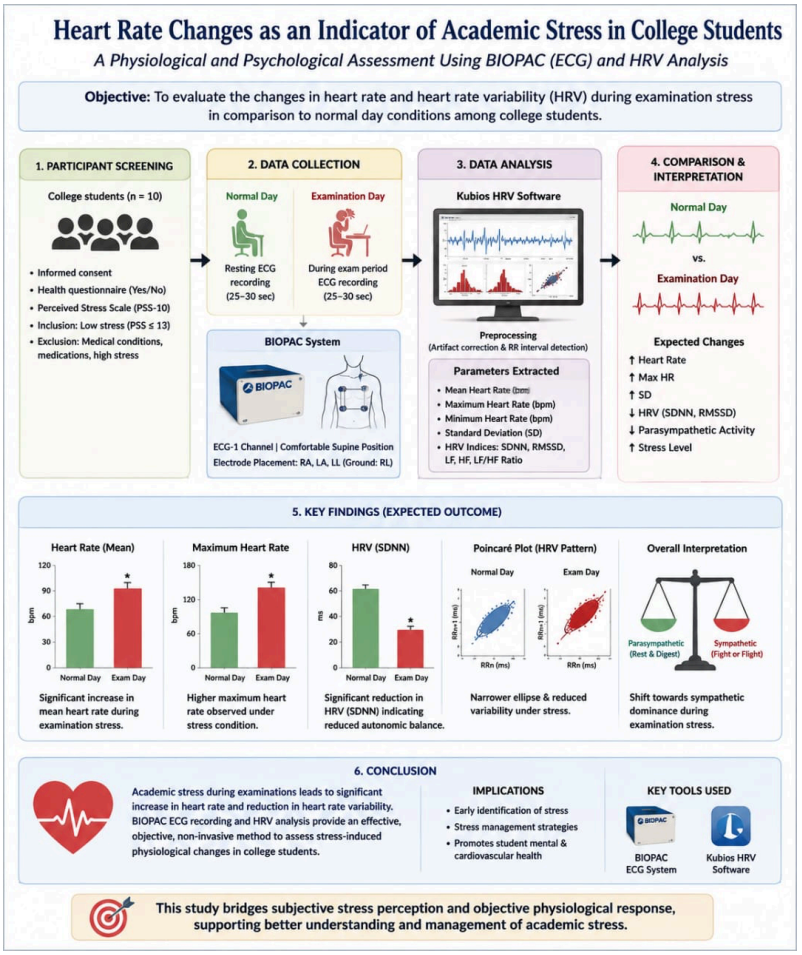


Figure:Graphical Abstract of ECG and HRV-Based Academic Stress Assessment in College Students

CHAPTER 1

INTRODUCTION

1.1 Academic Stress

Stress is a physiological and psychological response that is triggered when individuals are faced with a situation, or circumstances, that is challenging or demanding. One of the major stressors for students in higher education institutions is academic stress. Examination pressure, deadlines for assignments, competition, academic workload, and future career anxieties play a major role in students' stress.

Stress at school impacts on emotional health, learning capabilities, and body systems. Stress can have a positive impact on alertness and performance if there is a moderate level of stress, but a negative impact if there is too much stress on the individual in terms of their concentration, memory, emotional stability and physical health.
Students are under some of the most stressful conditions during exams. When students are tested, the physiological stress response systems such as increased activation of the sympathetic nervous system and the release of stress hormones are triggered, and thus facilitate fear of failure and performance anxiety.

1.2 Physiological Effects of Stress

Stress is a complex physiological and psychological reaction that is caused by a situation that the individual finds challenging, threatening, or demanding. Stress is a coordinated activation and activation of various biological systems in the human body to maintain internal homeostasis and to adapt to environmental changes. Stress at school, especially in examination periods, has the potential to have a profound effect on the function of the cardiovascular, endocrine, and autonomic nervous systems.
With stress, the body triggers a rapid sequence of physical reactions that are often summarized as “fight-or-flight.” This response is largely mediated by the ANS, particularly by the sympathetic part, which is in charge of reactivating the body very quickly in response to stressful stimuli. The stress response activates key physiological systems such as Autonomic nervous system (ANS), Endocrine system, Cardiovascular system.

The autonomic nervous system is important in controlling the involuntary functions of the body, including the heart rate, blood pressure, breathing, and glandular secretions. In stressful situations, increased activity of the sympathetic nervous system leads to greater physiological arousal. At the same time, relaxation and recovery responses are decreased due to a reduction in activity of the parasympathetic nervous system.
An activation of the sympathetic nervous system causes the adrenal medulla to secrete catecholamines like epinephrine (adrenaline) and norepinephrine (noradrenaline). These stress hormones have a number of immediate physiological effects such as An increase in the heart rate (HR), Elevation of blood pressure.

A rise in ventilation rate.A rise in the number of breaths taken per minute.Enhanced cardiac output: Pumping blood into the body for circulation increased glucose mobilization for energy production, Increased alertness and attention.

These changes are adaptive as they enable the body to act efficiently in stressful or challenging situations. But the constant or chronic activation of stress-response pathways can have detrimental effects on cardiovascular and psychological well-being.

In addition to autonomic activation, stress also stimulates the hypothalamic-pituitary-adrenal (HPA) axis. Activation of the HPA axis leads to secretion of cortisol from the adrenal cortex. The hormone cortisol has a role in regulating metabolism, modifying immune function and maintaining energy balance during stress. Persistent high cortisol can lead to feelings of tiredness, anxiety, insomnia, lack of focus, and heart and blood pressure issues.

The cardiovascular system is one of the first stress-related physiological systems to show evident and measurable changes. Under the stress, the activity of the sympathetic nervous system increases, which results in an increased activity of the heart and a measurable increase in the variability of heart rhythm and cardiac activity. Electrocardiography (ECG) and Heart Rate Variability (HRV) analysis can be objectively used to assess these changes.

HRV is the variation in the interval between each heartbeat and a representation of the balance between sympathetic and parasympathetic nervous system activity. Generally, HRV decreases and HR increases under stressful conditions, when sympathetic dominance occurs. Hence, HRV has been embraced as a significant non-invasive tool for autonomic nervous system (ANS) assessment and physiological stress response.
Students have been reported to be highly affected by academic stress particularly examination stress in several studies, which has been reported to significantly affect the functioning of the autonomic nervous system. In examination periods students may often suffer from anxiety, emotional pressure, fear of failing to do well and which all adds to a physiological stress response. Earlier study results have shown that examination stress conditions result in increased HR, decreased HRV, increased sympathetic activity and decreased parasympathetic activity in students.

Physiological responses to stress can be different for each person and depend on the level and length of stress exposure. Acute stress tends to cause autonomic changes that last for a short time; chronic stress can lead to chronic physiological responses that impact cardiovascular, endocrine, and psychological function.Physiological stress responses are assessed in the present study with the help of ECG-based heart rate monitoring and HRV analysis. ECG recordings can be taken using the BIOPAC MP36 system and offer objective cardiovascular data under normal resting conditions. ECG recordings and derived HRV parameters from them using the Kubios HRV Scientific Lite 4.3.0 software aid the evaluation of the autonomic nervous system balance and cardiovascular variability.

Therefore, it is important to have knowledge of the physiological responses to stress for assessing the effect of academic stress on cardiovascular responses and autonomic regulation in college students.

1.3 Heart Rate and Academic Stress

One of the key physiological parameters of cardiovascular activity and autonomic nervous system regulation is heart rate. There are several internal and external factors that affect it such as emotional state, physical activity, hormonal changes, environmental changes, and emotional stress. The autonomic nervous system is heavily altered during stressful periods, especially prior to exams, and this change is directly related to cardiac activity.

The stress of the academic environment triggers the sympathetic part of the autonomic nervous system, which raises the levels of catecholamines (adrenaline and noradrenaline). These stress hormones act on the sinoatrial node of the heart, increasing the frequency of heartbeat and the amount of blood pumped out. At the same time, parasympathetic activity is lowered, which diminishes the body's relaxation response, and increases physiological arousal.

Several investigations have shown that examination stress has a definite effect on the heart rate of students. Cardiovascular activation under examination is the result of the sum effect of increased mental pressure, fear of poor academic performance, competition and anxiety. Physiological adaptations are supposed to be responses to stress and a way for the body to cope.Because cardiovascular changes can be measured with a high degree of accuracy in real-time, heart rate monitoring is a common objective and non-invasive tool for assessing stress responses. HR assessment using ECG is a valid tool to monitor the functioning of the autonomic nervous system and stress regulation in the body.

Heart rate variability during stress also helps to assess autonomic balance, in addition to basic HR measurement. Sympathetic activation is the general trend with HRV decreasing when there is stress while the parasympathetic activity is decreased. Thus, the heart rate could be a useful indicator to be assessed when the academic stress is applied and it will be useful to understand the physiological effect of the academic stress in college students.

In the present study, ECG recordings and HRV analysis are used to assess the cardiovascular responses to academic stress. Physiological monitoring using an ECG is a scientific method to determine autonomic changes in students' body indicating stress.

1.4 Electrocardiography (ECG)

An electrocardiography, or ECG, is a widely-used non-invasive diagnostic and physiological monitoring technique that is used for the recording of the electrical activity produced in cardiac cycles of the heart. The electrical activity generated during the depolarization and repolarization of cardiac muscle cells is picked up by the electrodes in the body surface and illustrated graphically as ECG waves.

ECG is a useful tool in both clinical diagnosis and biomedical research, as it can provide information about the activity of the autonomic nervous system, electrical conduction pathways, cardiac rhythm and heart rate. ECG is one of the most widely used physiological evaluation methods in stress related and cardiovascular research, because it is simple, reliable and accurate.

The normal ECG waveform is made up of several characteristic components, such as the P wave, the QRS complex and the T wave. Every part is a particular electrical event in the heart.

P wave is associated with electrical activation of the atria (depolarization) before they contract. The QRS complex represents the depolarization of the ventricles, which represent electrical activity in the ventricles during contraction. T wave: this wave corresponds to the repolarization of the ventricles, that is, the return of the ventricles to their resting electrical state.

ECG recordings can be analysed to give detailed information about Heart rhythm and rhythm abnormalities, The heart's rate and activity, A heart tissue with the ability to conduct electricity.A heart tissue that can conduct electric current, RR interval variations Regulation of the autonomic nervous system, Physiological changes and cardiovascular changes are pathological.ECG is very significant for physiological stress research, as autonomic changes under stress conditions have a major impact on cardiac electrical activity. Stress-related changes in sympathetic/parasympathetic balance can be measured using ECG-derived parameters like heart rate and heart rate variability.

Different lead configurations are used for an ECG to record them for various purposes. For the present study, ECG recordings were obtained with the BIOPAC MP36 physiological data acquisition system (PDA) with a modified Lead II electrode configuration. The electrodes were placed in standard anatomical locations to minimize recording artifacts and record the signal accurately.ECG signals recorded during the recording were further analyzed by using Kubios HRV Scientific Lite 4.3.0 software to extract HRV parameters and to assess the activity of the autonomic nervous system.

Therefore, electrocardiography can be used as an effective and reliable physiological assessment tool to assess cardiovascular responses to academic stress among college students.

1.5 Heart Rate Variability (HRV) occurs

Heart rate variability (HRV) is the variation that occurs naturally between successive heartbeats. These variations are measured by the RR intervals taken from electrocardiographic recordings. HRV is considered an important physiological measure of autonomic nervous system regulation and cardiovascular adaptability.

Normally the heart doesn't beat in a perfectly regular rhythm. Rather, there is dynamic interplay between the sympathetic and parasympathetic branches of the autonomic nervous system which result in continuous fluctuations between successive heartbeats. A healthy autonomic nervous system keeps heart activity in balance so that there is normal variability of the heart rate.
Emotional, physiological and environmental changes have a strong impact on heart rate variability. Stressful situations typically cause an increase in the activity of the sympathetic nervous system and a decrease in the activity of the parasympathetic nervous system, which results in a decrease in HRV.

So, lower HRV levels are generally linked to stress, anxiety, autonomic imbalance, and sympathetic dominance.

Higher HRV, on the other hand, tends to signify good cardiovascular regulation and autonomic flexibility. HRV is a marker of physiological flexibility and greater stress resilience in people with higher HRV.HRV analysis can be done in the time domain and frequency domain. Time domain parameters measure statistical fluctuations in RR intervals as a function of time, while frequency domain parameters measure the distribution of autonomic activity as a function of frequency band.

In stress-related studies, common HRV parameters examined include:

1. SDNN (Standard Deviation of NN Intervals)
2. Overall heart rate variability is called SDNN, and represents the total autonomic nervous system activity. Lower SDNN is considered to be a sign of stress and sympathetic dominance.
3. RMSSD (Root Mean Square of Successive Differences)
4. RMSSD is mainly related to the activity of the parasympathetic nervous system and short-term changes in heart rate. Reduced RMSSD may reflect diminished vagal functioning in a stressful situation.
5. LF (Low Frequency)
   Low frequency components are affected by both the sympathetic and parasympathetic nervous system.
6. HF (High Frequency)
   The HF component is primarily due to parasympathetic regulation and respiratory-related cardiac variability.
7. LF/HF Ratio

It is seen that the LF/HF ratio is a good marker for autonomic balance. Increased LF/HF ratio may suggest sympathetic dominance during stressful conditions.

HRV analysis has emerged as a significant non-invasive tool to evaluate physiological stress responses, cardiovascular regulation, emotional states and autonomic nervous system functioning in recent years. HRV parameters are often used in academic stress studies to assess autonomic balance changes during exam sessions.

In this study, HRV parameters were obtained from ECG recordings with the help of HRV Scientific Lite 4.3.0 software. Cardiovascular variability and autonomic nervous system activity were assessed under resting conditions by performing analysis on RR interval data extracted from BIOPAC recordings of the ECG. Prior literature findings were used to interpret the HRV changes related to the examinations.

Therefore, HRV analysis is useful to objectively evaluate stress-induced autonomic changes from a physiological perspective in college students.

1.6 Aim of the Study

The present study is designed to investigate the ECG and HRV parameters to reveal the changes in heart rate and heart rate variability as physiological markers of academic stress among the college students.

The study emphasizes the understanding of the effect of academic and examination-related stress on ANS activity and cardiovascular regulation. Using the BIOPAC MP36 physiological data acquisition system, ECG recordings are made to evaluate the changes in HR and HRV variables in the resting state.

Further, this study will investigate the applicability of stress assessment using ECG based physiological monitoring and Kubios HRV analysis as objective and non-invasive tools in students. The results of the study can help to understand the cardiovascular responses to stress and the regulation of the autonomic nervous system in an academic stress situation.

1.7 Objectives

The major aim of the present study is as follows:

1. To acquire resting condition electrocardiographic (ECG) recordings from healthy college students with the BIOPAC MP36 physiological recording system.
2. Using Kubios HRV Scientific Lite 4.3.0 Software to analyze HRV parameters.
3. To derive physiological parameters from RR interval data obtained from the ECG, such as mean, minimum, maximum HR, standard deviation, SDNN, and RMSSD.
4. To assess autonomic nervous system function by HRV analysis and cardiovascular variability testing.
5. To make comparisons of predictions of physiological responses during examination stress conditions, from previously reported scientific literature.
6. To evaluate the effectiveness of ECG and HRV analysis as objective physiological means of evaluating academic stress in college students.

The degree of academic stress among college students causes certain physiological changes that affect the autonomic nervous system and cardiovascular regulation. ECG and HRV are reliable, objective and non-invasive means of measuring these physiological responses to stress. The purpose of the present study is to help shed light on the difference between academic stress and cardiovascular functioning, through the assessment of physiological variables by ECG and analysis of HRV. This study's results could also contribute to the use of physiological monitoring techniques in stress research and student health assessment.

CHAPTER 2

REVIEW OF LITERATURE

2.1 : ECG & HRV

Academic stress has become one of the most significant psychological and physiological issues of college and university students all over the world. The academic competition, examination pressure, workload and uncertainty of the future career are all factors that contribute to the sudden increase of stress among students. Research has clearly shown that academic stress impacts emotional and mental health, as well as causing measurable changes in the body's physiology, including those of the autonomic nervous system, the endocrine system and the cardiovascular system.

In recent years, increasing focus has focused on objective physiological measures of stress. Of these, electrocardiography (ECG) and heart rate variability (HRV) analysis has become increasingly important as reliable, non-invasive, and quantifiable measures of the activity of the autonomic nervous system and of cardiovascular regulation. HRV has been widely used in biomedical research to study cardiovascular adaptability, emotional regulation, autonomic balance, and stress responses, among others, and it has been extensively studied from the perspective of the ECG.

Scientific research literature reviewed here is related to academic stress, stress physiology, cardiovascular responses to stress, regulation of the ANS, electrocardiography, HRV, BIOPAC systems and Kubios HRV software. Past reported results are also reviewed for examination stress of students and stress assessment methods using physiological and psychological approaches are highlighted.The chapter theoretically and scientifically explains the physiological mechanisms that underlie stress-induced cardiovascular changes, thus creating a background for the present research.

2.2 Academic Stress among College Students

Academic stress is a physiological and psychological reaction that is created when academic demands are greater than an individual's adaptive ability. There are a number of stressors that constantly affect college students, such as tests, assignments, competition, peer pressure, lab work, oral presentations, deadlines, and future work plans. All these stressors combine to lead to higher rates of emotional and physiological stress.

The educational setting has become very competitive, increasing the number of stress-related conditions within schools. It is now acknowledged that academic stress is an important factor in public health, as chronic exposure to stress has adverse effects on cognitive functioning, emotional health, physical health, and quality of life.

While moderate stress can help improve concentration and motivation, too much or too long of a stress period can have a negative impact on learning ability, memory and decision making ability, and academic performance. Signs and symptoms of high academic stress include anxiety, irritability, sleep disturbances, fatigue, lack of concentration, and emotional exhaustion.

Study results suggest that academic stress can lead to Anxiety disorders, Depression, Sleep abnormalities, Reduced academic performance, Emotional instability, Burnout syndrome, Reduced social interaction, Cardiovascular disturbances.

Students find examinations one of the most stressful experiences as it consists of fear of failure, performance pressure and uncertainty about results. Examination stress triggers physiological stress response systems and triggers autonomic and cardiovascular changes that can be measured.

Many research studies have documented heightened levels of sympathetic nervous system (SNS) activity during exams. Stressful academic situations are often associated with increased heart rate, increased blood pressure, changes in respiration and decreased heart rate variability. These physiological responses reflect activation of the ANS and cardiovascular responses to stress.

The psychological stress and autonomic imbalance during exams has been consistently demonstrated in studies performed on medical, engineering and university students. It has also been found that there is research evidence suggesting that females may experience greater perceived stress than males; however, with physiological levels, they may be dependent upon their coping mechanisms and environment.

Academic stress can lead to stress-related health issues, not only in the short-term, but also in the long term, if the exposure to stress is chronic. Thus, it is essential to measure students' objective physiological stress responses to grasp the health impact of academic stress.

2.3 Stress Physiology

Stress physiology is a complex process of interactions between nerve, endocrine and cardiovascular systems, which play a role in preparing a body to cope with stress. When a person is exposed to a stressor, the body activates a series of physiological responses to stabilize internal systems and also prepares for fast reactions to the stressor.

The two main systems that are engaged in stress response:

1. The autonomic nervous system (ANS):
2. Hypothalamic-pituitary-adrenal (HPA) axis

The autonomic nervous system controls the unconsciously performed physiological functions such as respiration, digestion, glandular activity, heart rate and blood pressure. In times of stress, the sympathetic nervous system is activated and the parasympathetic nervous system is inhibited.

When the sympathetic nervous system is activated, these hormones are released from the adrenal medulla, including adrenaline and noradrenaline. The hormones have a quick effect on the body, such as Increased heart rate, Elevated blood pressure, respiratory rate, cardiac output, Release of Glucose from storage and awareness and alertness.
The responses are collectively named the "fight-or-flight response" and were first described by Walter Cannon. This response is meant to get the body ready for quick response in situations that are viewed as a threat or a challenge.

Stress activates the hypothalamic-pituitary-adrenal axis as well as sympathetic activation. A hormone called corticotropin-releasing hormone (CRH) is produced in the hypothalamus and stimulates the release of another hormone, adrenocorticotropic hormone (ACTH) from the pituitary gland. ACTH then promotes the release of cortisol from the adrenal cortex.Cortisol is an important glucocorticoid hormone involved in Regulation of metabolism, Steering clear of high blood sugar.Avoidance of hyperglycaemia, Immune system modulation, Relaxed, alert and ready to respond to a threat. Ability to mobilize energy during stress.
While acute exposure to cortisol is a normal response, excess cortisol can have negative impacts on cardiovascular function, cognitive abilities, immune function, and emotional health.Autonomic activation is a stress-dependent phenomenon that is a strong determinant of cardiovascular physiology. The more sympathetic activity there is, the more cardiac activity there is, and the less parasympathetic activity, the less cardiac relaxation there is. These autonomic changes are manifested as changes in heart rate and heart rate variability.

Cardiovascular and autonomic control are therefore tightly coupled with the physiology of stress, so that the ECG and HRV analysis can be useful in the physiological evaluation of stress.

2.5 Autonomic Nervous System

The ANS is a physiological system that is important in the control of involuntary functions essential for internal homeostasis. It regulates several body functions such as breathing, blood pressure, digestion, heart rate, glandular activity, thermoregulation and respiration. Physiological activity is constantly modified by the autonomic nervous system in response to the environment and emotional state.

The autonomic nervous system is largely composed of two functional divisions – sympathetic nervous system and parasympathetic nervous system. These divisions have opposite but complementary effects on the physiology of different organs and systems, which result in physiological homeostasis and balance in the body.

The sympathetic nervous system is activated when it's stressed or demanding. It acts to mobilise the body for quick response, by stimulating physiological activity and energy mobilization. Activation of the sympathetic system causes the adrenal medulla to secretes adrenaline and noradrenaline, thereby increasing the activity of the cardiovascular and metabolic systems. Sympathetic stimulation causes an increase in heart rate, blood pressure, respiration rate and alertness.

The parasympathetic nervous system, on the other hand, is responsible for relaxation, recovery, digestion and conserving energy. The vagus nerve is the main mediator of parasympathetic activity and aids in cardiovascular stability in resting conditions. A rise in the parasympathetic activity decreases the heart rate and facilitates physiological recovery from a stressor. In a healthy state the two systems of nerves—sympathetic and parasympathetic—are in a state of dynamic equilibrium. This balance enables the body to adjust well to the demands of the environment and body. Stressful situations, on the other hand, throw off autonomic balance, with an increase in sympathetic dominance and decrease in parasympathetic activity.

Autonomic imbalance is strongly associated with stress-related physiological changes. During stress, increased SNS activity increases cardiovascular excitation, and decreased activity of the PSNS decreases autonomic flexibility. These changes are manifested as measurable changes in heart rate and heart rate variability.

Heart rate variability (HRV) is one of the most reliable measurements of autonomic nervous system (ANS) function, as it is a measure of beat-to-beat fluctuations in cardiac rhythm that are linked to ANS control. In general, high HRV is a positive sign of healthy autonomic adaptability and effective parasympathetic functioning, while low HRV is a sign of stress, anxiety, fatigue, emotional exhaustion, and sympathetic dominance.There has been a number of research studies which have been able to show that examination stress has a significant impact on autonomic nervous system activity among students. In stressful school environments, higher HR and lower indices of HRV are often seen due to higher levels of sympathetic activation and lower parasympathetic modulation. This indicates that during stress exposure, there is a disturbance in the autonomic balance.

The endocrine system also plays a close relationship with the autonomic nervous system in stress reactions. Cortisol is released from the HPA axis and acts as a modulator of neuroendocrine function, further impacting autonomic activity upon activation. This interaction helps to maintain physiological activation over the long term during prolonged stress.

Chronic stress and its associated autonomic dysfunction can be a factor in a variety of health problems such as cardiovascular disease, high blood pressure, sleep disorders, emotional disorders and anxiety. Hence, autonomic function is increasingly recognized as an important aspect to study in stress physiology.There are modern methods of physiological monitoring that enable objective assessment of autonomic activity based on ECG-derived HRV analysis. SDNN, RMSSD, LF power, HF power, and LF/HF ratio are parameters that are important to have information about sympathetic and parasympathetic regulation.

The current research adopts the ECG and HRV analysis to assess the ANS response of college students under the conditions of academic stress. Learning about autonomic regulation during stress could help to advance the physiological monitoring of student health and emotional well-being.

2.6 Electrocardiography (ECG)

Electrocardiography is one of the most common non-invasive methods of recording electrical activity of the heart. ECG is now an invaluable instrument to clinical medicine, cardiovascular physiology, biomedical engineering and stress related physiological research, as it allows accurate and continuous evaluation of cardiac electrical activity.

The heart produces electrical charges in special conducting tissues which coordinate the heart muscles to contract properly. Electrodes that are placed at certain anatomical sites can detect these electrical signals from the body surface. The electrical activity recorded is displayed graphically as an electrocardiogram.

The ECG trace has several important features such as the P wave, the QRS complex and the T wave. Each component is a specific aspect of cardiac electrical conduction and cardiac myocardial activity. The P wave is depolarization of the atria, the QRS is depolarization of the ventricles and the T wave is repolarization of the ventricles.

The electrocardiogram is helpful in the assessment of heart rhythm, cardiac conduction, heart rate, arrhythmia, myocardial abnormalities, and autonomic nervous system activity. ECG has become an important method to assess physiological stress as the autonomic control directly affects cardiac rhythm.The recording of the ECG is especially important in stress research since emotional and psychological stress have a highly significant influence on cardiovascular regulation. Under stress conditions, the pattern of the heart rate (HR) changes along with the RR interval, and the changes in those intervals can be quantitatively measured using the electrocardiogram (ECG) and HRV.

ECG is very well suited for physiological monitoring for several reasons. It is non-invasive, relatively inexpensive, accurate, reproducible and can be done in real time. In the modern ECG systems, digital signals are also acquired, and cardiac parameters are also analyzed in a digital computer.ECG recording systems like BIOPAC are commonly used in biomedical research labs to record physiological signals. These are the systems that would give high-quality recordings with negligible distortion in the signal and would even enable the monitoring of multiple physiological parameters simultaneously.

In the ECG recording, electrodes are placed on certain points of the body to monitor the electric potentials produced by the cardiac activity. The electrode placement used in this current study is similar to that of a modified Lead II configuration, with the addition of a right arm electrode. To get accurate ECG waveforms and to reduce recording artifacts, the right placement of electrodes is important.Analysis of RR interval in an ECG recording can be used to give useful information on autonomic nervous system activity. Heart rate variability analysis is based on the variability of RR intervals, which is a measure of autonomic modulation of heart rhythm.

A number of studies have been conducted with students and have shown that there are obvious changes in the ECG under examination stress conditions. Three consistent findings have been observed during stressful academic situations: increased heart rate, decreased RR interval variability, and sympathetic dominance. The electro-physiological results show that ECGs are very useful for objective stress evaluation.

In recent years, there has been a new advancement in digital signal processing and physiological software, which have further enhanced the ECG analysis techniques. Advanced analysis of time-domain, frequency-domain and nonlinear HRV parameters from ECG recordings is possible with modern HRV software like Kubios.

The research of the present study involves the use of BIOPAC systems to record ECG data that is then analyzed with Kubios HRV software to assess ANS activity and physiological responses to stress in college students. Physiological monitoring using ECGs can thus be seen as a sound scientific tool to gain insight into cardiovascular changes under stress.

2.7 ECG in Stress Research

The development of electrocardiography has made it one of the most widely applied physiological measures in stress-related biomedical research, since the nervous system's autonomic and cardiac electrical control mechanism is sensitive to psychological and emotional stress. With ECG, objective measurements of cardiovascular reactions occurring during a stress experience are obtained through a non-invasive means. Because of its accuracy, reliability, and its capacity to be continuously monitored in real-time, it is utilized in a variety of settings, including the clinical area, neuroscience, behavioral physiology and psychophysiology.

Over recent years, considerable research has emerged investigating the association between psychological stress and cardiovascular activity. Various stress models including, academic stress, occupational stress, emotional stress, mental workload, anxiety disorders have been studied using ECG-based physiological monitoring methods. Stress related to examination seems to be one of the commonly used and powerful stress paradigm applied in studies with college students, because examination is considered to be one of the major sources of stress to a young individual.

Stress-induced physiological changes are indicated by deviations in ECG waveforms and pattern of the RR intervals. During a stressful episode the heart rhythm and cardiovascular parameters are regulated by the increased activity of the sympathetic nervous system which has an effect of accelerating heart rate and changing the rate and timing of cardiac electrical activation. Therefore, increased sympathetic and reduced parasympathetic activities during stress can be observed as shortened RR intervals, enhanced cardiac excitation and autonomic control.Reduced parasympathetic activation causes reduced HRV.

It has been shown in a few student-based studies that examinations cause various changes to cardiovascular parameters monitored by ECG. Enhanced heart rate during examination period is one of the most consistent physiological changes found during academic stress research. Examination has been indicated to evoke anticipation stress in students which stimulates sympathetic activity before the examination event actually takes place.

Studies on medical and engineering students showed that vagal tone was significantly reduced and HRV indices lowered during examination stress, accompanied by elevated sympathetic activity in relation to examinations. Autonomic changes are associated with reduced ability to relax and heightened cardiovascular arousal. Occupational physiology has also utilized ECG in the studies. High performance conditions, continuous mental workload, duty work and emergencies resulted in altered autonomic activity which is indicated by ECG. High heart rate, low HRV has been indicated as correlated with emotional exhaustion, fatigue, anxiety, poor psychological adaptability to the stressor and low coping potential.

Emotional stress, including the examination of fear, anger, sadness, social pressure, and emotional conflict, can be measured through ECG for its effect on the cardiovascular system. Since the limbic and hypothalamus have profound effects on the cardiovascular system, emotional stress strongly influences the regulation of the cardiac rhythm and the balance between autonomic systems, and thus leads to heart rate acceleration and altered RR interval patterns. Anxiety disorders and panic disorders are also often associated with significantly high heart rate, low HRV and impaired vagal tone during resting conditions.

The main reason for utilizing ECG during stress studies is to obtain an objective measurement rather than relying solely on the verbal and psychological questionnaires of the subjective feelings, as emotions may differ in individuals with different personality structures. With computerized systems now available, digital signal recording, artifact rejection and RR interval data analysis as well as modern HRV analysis can be done accurately and easily. BIOPAC recording can provide reliable ECG for physiological research with subjects being either health individuals or patients.

Heart Rate Variability (HRV) is also increasingly employed with ECG to obtain the information on the autonomic nervous system. The measurement of various time-domain and frequency-domain indices of HRV from ECG signals is widely used as reliable indices of vagal and sympathetic activity.

In the present study, ECG measurement will be taken as a measure for physiological indicator and analysis will be done on its characteristics as well as HRV features that are expected to reveal changes in cardiovascular control mechanisms induced by academic stress in college students. These indices will be compared between normal condition and condition of expecting examination stress.

2.8 Heart Rate Variability (HRV)

Heart rate variability represents the natural variability of time interval between consecutive heart beats. HRV occurs due to the constant interaction between sympathetic and parasympathetic branches of the autonomic nervous system, and it can be considered one of the most important physiological measures of autonomic regulation and cardiovascular plasticity.

Although heart beats appear rhythmic, the interval between two heart beats is not uniform; a continuous interplay between sympathetic and parasympathetic mechanisms is needed to adjust the activity of the cardiovascular system according to both internal physiological requirements and external environmental factors. So, heart rhythm variations indicate activity of the autonomic nervous system.HRV measurements can be taken by RR intervals derived from an ECG trace. The RR interval is the duration of time between two consecutive R peaks on an ECG signal, which can give an idea about the activity of the autonomic nervous system and of the cardiovascular regulation.

A high HRV score indicates autonomic balance, good parasympathetic activity and high adaptability. Reduced HRV is generally associated with stress, anxiety, fatigue, burnout, autonomic dysfunction and cardiovascular risk.

HRV measurement is widely used in biomedical research due to its accuracy in reflecting autonomic nervous system function objectively. Current HRV analysis has two branches, time domain analysis and frequency domain analysis, plus non-linear measures which represent the complexity of physiological control systems.Time-domain parameters, such as standard deviation of normal RR intervals (SDNN) and root mean square of successive RR interval differences (RMSSD), assess time-course variation. SDNN provides information on the overall variation of autonomic input, whereas RMSSD is closely linked to parasympathetic control.

Frequency domain parameters, including low frequency (LF) and high frequency (HF), indicate the activity of the autonomic nervous system across the different frequencies. HF is regarded as parasympathetic nervous system control of the heart, whereas LF indicates both sympathetic and parasympathetic control and the LF/HF ratio is used as an indicator of the sympathovagal balance.
Non-linear parameters, like poincar plots, entropy analysis, and fractal measures are more complex indicators which demonstrate the system's adaptability to challenges and can reflect how the physiological system adapts to changing circumstances, like stress.

HRV changes during stressful situations, as activity of the autonomic system will rapidly fluctuate. A decrease in HRV can indicate loss of balance, in which the parasympathetic activity reduces and sympathetic activity increases, affecting the cardiovascular plasticity. Various studies demonstrated the correlation between acute stress and loss of vagal activity and therefore HRV reduction.

Academic stress researchers among students are able to link the level of examination stress with the reduction in HRV during the examinations. Students have higher sympathetic dominance and decreased parasympathetic activity during examinations, which will cause SDNN and RMSSD values to be low, which reflect a state of physiological stress. HRV is reduced also in various mental states and pathologies, such as anxiety disorders, depression, sleep disorders, chronic fatigue and diseases of the cardiovascular system; this information on the autonomic nervous system might also tell us about the health status of the heart.

It has been only recently with computerised ECG measurements and specialized HRV analysis software, such as Kubios HRV software, that research on HRV can be conducted efficiently and reliably, by analyzing all of the various components of HRV. In this research, a time-domain analysis and a frequency-domain analysis will be used with measures of SDNN, RMSSD, LF, HF andLF/HFratio.

2.9 HRV and examination stress

Examination stress has one of the most common stress types analyzed by psychophysiologists, since examination implies cognitive load, emotional arousal and physiological pressure for students. Numerous previous studies have confirmed that examination stress affects the autonomic regulation system as well as cardiovascular functioning.

Heart rate variability is recognized as an indicator for autonomics and a strong measure for monitoring the body's physiological responses to stressors like examinations. Stressful circumstances like examinations increase sympathetic activation, while simultaneously decrease the parasympathetic control. Such changes can be identified through measuring changes in HRV values.

Consistently, HRV is shown to decrease during examinations and an important factor of physiological stress is an increased lack of physiological adaptability. Multiple university students were found to have less parasympathetic activation while demonstrating higher heart rate and sympathetic control during exams. One of the psychological aspects leading to autonomic activity during examination stress is anticipated fear. Students often encounter pressure related to performance, potential failures and cognitive overload which activate their autonomic system, increasing sympathetic drive and variability in the RR intervals.

Ecg and HRV measures taken during examinations have revealed decreased parasympathetic activity shown by decreases in SDNN and RMSSD values. Increase in LF/HF ratio during stressful conditions suggest an increase in sympathetic activity relative to parasympathetic activity.

Reduced emotional control, decreased sleep and concentration, and decreased cognitive functioning also have been reported.Individual differences in individual ability to cope with the stress, individual personality and knowledge are responsible for variation in these measures.

Medical students have had studies performed on them, in which they displayed various types of autonomic changes during exams as they tend to work in environments with a very high amount of stress due to their study program. These results are similar to those found in studies of other students, such as those studying engineering or university sciences.

The clinical importance of the reduced HRV during stressors is that chronic dysregulation will negatively influence the cardiovascular system. Chronically increased sympathetic activity and reduced vagal tone may cause problems like increased blood pressure, anxiety disorders, burnouts, poor recuperation capacity and so on. An autonomic system measured using the HRV allows an unbiased objective measure of the stress experienced by a person, without relying on subjectively determined values from questionnaires alone, thus giving it a high level of reliability as a measure.

An ECG derived HRV will be used in this research to quantify the level of autonomics present during academic stress in a group of college students, and measures of SDNN, RMSSD, LF, HF and LF/HF ratio will be analysed with the help of Kubios HRV software.

Besides autonomic changes, researchers found that examination stress could impair sleep, emotion regulation, attention and cognitive functioning. The degree of autonomics variation under examinations depends on various factors like individual characteristics, coping mechanisms, personal attitude and supportive environment.

Medical students' autonomics responses have been reported to change during examination sessions due to high workload and mental pressure from studying and performing during their education. Similar autonomic responses were recorded among students in other fields of study like engineering and sciences, thus showing universality of these findings. Reduced HRV is an important clinical measure because excessive stress may have adverse consequences for long-term cardiovascular health. Excessive prolonged high sympathetic stimulation coupled with a lack of parasympathetic activation might be a risk factor for cardiovascular disease and various other disorders such as anxiety, burnout etc.

HRV represents a good method to objectively study stress as these objective physiological measures will not depend on subjectively interpreted responses from questionnaire measures. Coupling both measures would create a reliable stress measure system.

ECG derived HRV measures will be used in this study, however examination day could not be assessed due to limitations in equipment from BIOPAC, but a review of possible answers were presented based on previous results on HRV and stress.

2.10 BIOPAC System in Physiological Research

The BIOPAC system has been widely used in biomedical engineering, physiology, neuroscience, psychology and medical laboratories for the recording and analysis of physiological signals. BIOPAC provides real-time accurate, and reliable physiological recording systems for use in both clinical and research settings.

The BIOPAC data acquisition system is capable of recording and measuring a range of physiological signals such as electrocardiogram (ECG), electroencephalogram (EEG), electromyogram (EMG), respiration, blood pressure, galvanic skin response (GSR) and pulse. Due to its precision and functionality BIOPAC is one of the most commonly used physiological recording systems in university and biomedical research laboratories.In cardiovascular physiology research the BIOPAC system has been extensively used for ECG recording and HRV analysis. The accuracy of RR intervals measurement and consequently autonomic function, depend heavily on high quality ECG recording. BIOPAC systems are designed for the consistent recording of signals with negligible electrical noise and interference and absence of artifacts.

The BIOPAC MP36 system, which is widely used in educational and biomedical laboratories, consists of a multi-channel physiological recording system, digital amplifier, computer based analysis and real time physiological plotting display. It allows for accurate calculation of heart rate, RR intervals, waveform shape and cardiovascular response.This system is also highly convenient because it is capable of transferring physiological recordings to analysis software packages. EC G recordings acquired using the BIOPAC system may be transferred to analysis software programs such as Kubios HRV for analysis.

During ECG recording electrodes are attached to specified locations on the body following specific lead configurations, which may impact the accuracy of ECG waveform and noise artifacts. In this study a modified Lead II electrode placement configuration was used comprising right arm, left arm, left leg and earth.

BIOPAC systems are also extremely useful for students and it helps in providing practical knowledge about physiological signal acquisition and cardiovascular physiology. Practical training with BIOPAC systems enables students to gain an in-depth understanding of ECG recording procedures, electrode placements and signal interpretation as well as physiological analysis.

A number of studies have been carried out to investigate the effects of stress physiology using the BIOPAC system for ECG based autonomic assessment. Stressors such as examination stress, emotional stress, cognitive workload and anxiety based physiological parameters have been successfully analyzed using BIOPAC system ECG recordings.

The present study will be using the BIOPAC MP36 system to measure ECG recordings of students under resting state. The measurements obtained will be analyzed for heart rate and autonomic nervous system activity using the Kubios HRV software.

2.11 Kubios HRV Software

The Kubios HRV software is among the most widely used software program for heart rate variability analysis in scientific context. Its use has spread widely in biomedical engineering, physiology, cardiology, neuroscience, sports science and psychophysiological research due to its capability to provide an accurate and detailed analysis of autonomic function.

Kubios HRV software has the ability to accurately analyze RR intervals extracted from ECG recordings. It possesses advanced features such as automated artifact correction, preprocessing of RR intervals, time-domain analysis, frequency-domain analysis and non-linear analysis of HRV, trend correction and plotting. One of the most important features of Kubios HRV software is its ability to detect accurately and remove artifacts from recordings. Recording systems are susceptible to the effects of movement artifacts, electrical interference, and noise which may affect accurate calculation of HRV parameters. It provides automatic or manual artifact removal capabilities.

Time-domain analysis of HRV data includes variables such as mean RR interval, mean heart rate, SDNN, RMSSD and heart rate standard deviation; which are calculated by Kubios to demonstrate variability in the time domain. These variables may reflect general autonomic variability and parasympathetic control.

Frequency-domain analysis is used to assess autonomic control in different frequency bands and is performed using Kubios HRV. It allows calculation of total power, low frequency power, high frequency power and the low frequency to high frequency power ratio (LF/HF ratio) that are used to depict balance between sympathetic and parasympathetic control over cardiovascular function.

Kubios software also provides advanced non-linear analysis of HRV, such as Poincare plots. Poincare plots illustrate the variability of the RR interval data, providing a visual representation of autonomic function.

Numerous studies evaluating stress physiology have successfully used the Kubios HRV software to provide an objective measure of autonomic nervous system activity. It is well established that during examinations HRV measures decrease, and sympathovagal imbalance increase as detected using Kubios HRV software.It has gained significant importance within the field of psychophysiological research, as there are strong associations between autonomic nervous system activity and emotions regulation, stress, and cardiovascular response; information which can be gathered from Kubios HRV software analysis.

The current study will be using Kubios HRV Scientific Lite 4.3.0 for analysis of the RR interval extracted from the ECG recorded using the BIOPAC system. Several parameters will be calculated, including mean heart rate, min heart rate, max heart rate, SDNN, RMSSD, LF, HF and the LF/HF ratio; for measurement of autonomic nervous system activity.

2.12 Previous Research Studies

There is a considerable number of studies that have examined the correlation between stress of exams, autonomic nervous system control and cardiovascular function among college students. Each study concludes that examination stress alters heart rate and other HRV parameters.

The increase of heart rate and the subsequent decrease in HRV during exams implies greater sympathetic control and reduced parasympathetic modulation which reflects lack of autonomic balance.

There is evidence that studies involving students in medical school consistently record a substantial increase in stress response because medical education carries high academic demand and emotional burden.Similar studies conducted in nursing school, engineering school, and other university student cohort also reveal that stress alters their autonomic nervous system function.

There are a few studies that utilized ECG and HRV to demonstrate the measurable effects of stress using non-invasive physiological recordings. These studies demonstrated an increase in heart rate and decreases in the variables SDNN and RMSSD as well as elevation in LF/HF ratio during examination stress.

Several research studies have combined subjective measurements of stress (e.g., Perceived Stress Scale questionnaire) with physiological measurements (ECG, HRV) for an increase in the reliability of measures in studying stress responses among students.

The evidence supports the utilization of ECG and HRV analysis in stress physiology research due to its accuracy and non-invasive approach in assessing autonomic nervous system function.

2.13. Research Gap

Despite research examining academic stress and the autonomic nervous system, a limited amount of work has been carried out examining academic stress by combined assessment of EC, HR V analysis, and psychological questionnaires in Indian college students.

Previous works are largely either looking at physiological parameters only, or questionnaire approaches. Very few researches combined both physiological assessment with questionnaire to find stress. In addition, within the Indian context, only a handful of studies investigated ECG-based HRV analysis in undergraduate students by using BIOPAC system and Kubios HRV analysis system. India has different types of academic pressure, cultural factors, study pressure, and daily routine, so the physiological aspect of stress might differ in the Indian context.

Lastly, another weakness was the relatively small number of studies in which ECG recording training with real time physiological data analysis in students. Investigations combining physiology education, ECG recording, HRV analysis and stress evaluation are restricted. In the present study, we aim to contribute towards this research gap by evaluating the academic stress among students with combined ECG recording, HRV analysis using Kubio's analysis system, and psychological stress assessment, emphasizing physiological monitoring and objective evaluation of ANS activity in response to stressful conditions.

CHAPTER 3
MATERIALS AND METHODOLOGY

3.1 Study design

The present study was carried out as an observational, comparative physiological research to assess cardiovascular responses during academic stress in college students. This study involved an investigation on changes in heart rate and autonomic function by employing ECG and HRV.

Non-invasive physiological recording was carried out using the BIOPAC MP36 data acquisition system and Kubios HRV Scientific Lite 4.3.0 software.ECG recordings from the resting conditions were taken to identify heart rate and HRV parameters. The examination stress response was investigated by referencing from previously reported literature due to technical failure of the BIOPAC system that prevents recordings during the actual examination-day; thus the response was modeled by estimation from literature review.In general the investigation was aimed to establish a relation between cardiovascular physiology and academic stress by objective physiological measurements. Due to the influence of academic stress on autonomic system balance,ECG and HRV measures were selected as reliable indicators of stress.

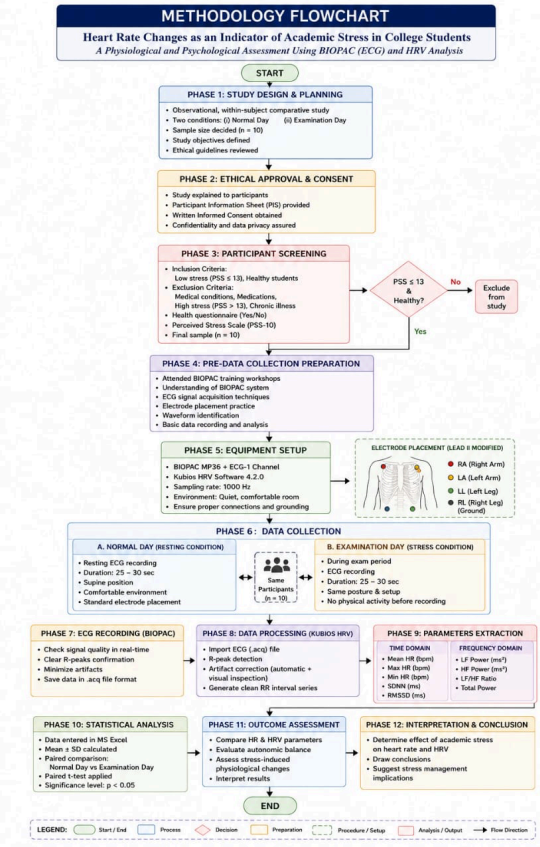


Figure3.1.1:Methodology Flowchart of ECG and HRV Analysis Study

Measurements were taken in the laboratory setting in order to reduce influence of uncontrollable environment variables on recorded parameters. The physiological recording and measurement procedures were kept consistent for all subjects in order to minimize any inconsistencies.

3.2 Participant recruitment

Selected healthy undergraduate students volunteered for this physiological investigation. Students were chosen as subjects for investigation since the cardiovascular system has great impact from academic stress related to examination, assignment, and workload as it is very common among university students.

All volunteers for the investigation were provided with the brief explanation on objectives and procedure of the study prior to the start of the measurement. Students were also informed about the ECG recording, HRV analysis, and confidentiality and privacy of their physiological data were guaranteed.

Health screening for each volunteer was conducted prior to data acquisition. They were required to complete a health questionnaire and Perceived Stress Scale(PSS-10) questionnaire before inclusion to the research. A total of eleven volunteers completed the initial screening and were further analysed according to inclusion and exclusion criteria after which one student was ruled out as they have health related issue that might affect cardiovascular parameters. So, a total of 10 students participated in the final study.

Participants chosen for the study are all physically fit and have normal physiological state appropriate for cardiovascular and autonomic nervous system examination. The subject's cooperation was very vital to ensure good quality physiological recording.

3.3 Inclusion criteria

Inclusion criteria was developed to ensure homogeneity of subjects in this study. Students with absolutely normal physiological systems, no diagnosed cardiovascular disease and no previous chronic medical condition that might affect the autonomic nervous system state, were included. Only students with PSS scores 13 or less were selected to keep cardiovascular responses to be relatively stable by selection from low stress groups. Besides, a student who agrees to participate in ECG recording procedure and HRV analysis was included. Also physically comfortable and mentally alert participants, those who were able to stay still during recording procedure, were included to obtain accurate measurement of cardiovascular parameters.

3.4 Exclusion criteria

Subjects having cardiovascular disease history, such as heart arrhythmia, hypertension, coronary heart disease, congenital cardiac disease, were excluded because these health conditions can directly affect cardiovascular regulation.

Participants having chronic illnesses such as diabetes mellitus, respiratory illnesses, neurologic disorder, endocrine condition and any form of autonomic nervous system imbalance were excluded since the symptoms of the mentioned diseases can also directly affect the measurements performed in this study. Also, subjects consuming medication that has been known to cause physiological changes on the cardiovascular system such as rate of heart beat and blood pressure were excluded as it may interfere with cardiovascular measure. Psychologically highly distressed students with high level of stress and severe anxiety are also excluded to ensure the data accuracy from physiological influence factors to the measurements.

3.5 Ethical Considerations

Ethical principles and patient safety were maintained at the highest possible standard. Participants were provided with clear explanations regarding the purpose, procedures and expected outcomes of the study prior to commencement of the data collection process. All participants were advised of the techniques that would be used for recording of ECG signals and analysis of heart rate variability. Participants were notified that the methods would involve non-invasive physiological recording without the use of drugs or invasive techniques and written informed consent was obtained from each participant prior to the commencement of the study.

The right to withdraw from the study at any time without academic or personal penalty was emphasized. Participants were assured that the information obtained during physiological recording would remain strictly confidential and would only be used for academic purposes, and that all physiological data would be discarded after completion of analysis. Physiological recording was conducted in a safe and comfortable laboratory environment and rest periods were encouraged prior to ECG acquisition to stabilize cardiac activity.

All physiological recordings were performed in accordance with the current guidelines regarding ethical considerations in biomedical and physiological research involving human participants.

3.6 BIOPAC System

The BIOPAC MP36 data acquisition system was employed to record physiological parameters for the present investigation. The BIOPAC systems are extensively utilized in medical physiology laboratories, neuroscience laboratories, biomedical engineering laboratories, and clinical research laboratories as these systems are reliable and accurate when used for acquiring physiological signals from the human subjects.

The BIOPAC system is a tool that helps record, amplify, process and analyze physiological data recorded from the human body. Using this system it is possible to record cardiovascular activity, muscle activity, respiration, and neurological activity from the body simultaneously, in a controlled environment.

During the current research work, the BIOPAC was predominantly utilized for the acquisition of ECG signals and analysis of heart rate. From these ECG recordings heart rate variability parameters and autonomic nervous system states were analyzed using Kubios HRV software.

The BIOPAC MP36 system has the sophisticated design of signal amplifiers, interface to electrodes, computerized data acquisition modules, and recording software to enable acquisition of physiological data. These systems are also capable of transferring the physiological signals from the analog form to the digital form suitable for computer-based analysis and graphical displays.
BIOPAC systems are highly beneficial as they are capable of acquiring high quality physiological signals and thus the level of interference or artifact in the recording should be minimum. For accurate ECG acquisition,RR interval determination and subsequent analysis of HRV measures; the accuracy and precision of the system are of high importance.
During this study the BIOPAC MP36 acquisition unit was used, which has several channels to monitor multiple physiological parameters concurrently. But here the ECG-1 channel was used to record the cardiovascular activity of the participants.

It also provides a real time waveform display for the ECG, and thus, the actual cardiac activity is recorded during the recording session, so that any recording artifacts or electrode issues can be immediately rectified.

BIOPAC systems are commonly used in biomedical education, to impart basic principles of physiology and signal processing. Medical, and biomedical engineering students can utilize this equipment to study parameters associated with cardiovascular physiology, neurophysiology, EMG, respiratory activity, and stress physiology.

The high reliability and accuracy of the BIOPAC systems make them an ideal system for studying physiological stress response as autonomic Nervous System activity and regulation of the heart function are significantly affected by stress, and hence ECG measurements are imperative.

Components of BIOPAC System

The BIOPAC system consists of various interconnected hardware and software components for acquisition and analysis of physiological signals.

The MP36 acquisition unit is the primary hardware component which amplifies and digitizes physiological signals from electrodes, converting them from analog electrical signals to digital computer signals which are able to be analyzed.
ECG amplifiers present within the BIOPAC system provide clear depiction and enhancement of the low amplitude cardiac electrical signals and their waveforms. As cardiac bioelectrical signals are weak naturally, they must be amplified by electrodes.

Surface electrodes are utilized in the determination of the electrical activity resulting from cardiac depolarization and repolarization and act as interfaces between the body and the recording equipment.

The BIOPAC acquisition software displays the recorded physiological signals in real-time for viewing, storing and recording, and provides the capability to analyze and export data for further analysis of HRV.

A computer interface is employed within the system to allow hardware and software to combine to create storage of digital signals for management of physiological data.

All of these components can be connected together to form a highly effective physiological acquisition system useful for biomedical research and cardiovascular assessment.

ECG-1 Channel

In the current study the ECG-1 channel on the BIOPAC MP36 system was utilized to record the cardiac electrical signal.

The ECG-1 channel of the BIOPAC MP36 allows the electrical activity associated with the depolarization and repolarization of the cardiac muscle to be displayed.

When the system is employed to record the cardiac electrical signal, leads are attached to the participant, to acquire electrical activity which are then displayed as ECG waveforms after amplification by the system.

The main electrical potentials seen on the ECG waveform include the P wave, QRS complex, and T wave. These electrical potentials correspond with atrial depolarization, ventricular depolarization and repolarization, respectively. The ECG-1 channel allows for clear viewing of RR intervals, crucial for heart rate variability (HRV) calculation, since parameters are derived from the variation between consecutive RR intervals.

As it yields clear visualization of RR intervals and is crucial for HRV, the ECG-1 channel was utilized for physiological recording for cardiovascular assessment in the present study.

Electrode Placement

Careful placement of ECG electrodes are necessary in order to prevent signal distortion. A modified Lead II was used because it displays a stable ECG tracing and theRR intervals, as it will be explained later.

The four electrodes were attached to the subject in specific regions: Right arm, left arm, left leg, and on the right leg for the ground electrode. The right arm electrode acts as the negative input, the left leg as the positive input, the left arm stabilizes the waveform, while the right leg is used as a ground electrode to eliminate electrical interference.

Before attaching electrodes to the skin surface, the areas where the electrodes were to be placed were cleaned to ensure a better electrical connection and to prevent noise artifacts by minimizing impedance. It is important to attach all electrodes in the precise anatomical location used for all participants, as it allows for consistent results to be obtained when recording ECGs from subjects.

Signal Acquisition

1. The BIOPAC system monitors physiological activity by transducing analog bioelectrical signals to digital signals for processing by a computer.
2. Electrical impulses from the heart are picked up by electrodes on the surface and transferred to the acquisition unit for amplification, filtering, and digitization, and then for visual representation.
3. It is necessary to filter these signals, as there will always be electrical interference from the environment or motion artifacts or baseline shift in any recording.
4. BIOPAC ECG signals are digital and thus can be stored, exported, and analyzed using specialized cardiovascular analysis programs such as Kubios HRV Scientific Lite.
5. The above sequence allows the measurement of physiological activity and cardiovascular function to be objectively determined.

BIOPAC advantages

1. The BIOPAC system has a number of advantages in terms of physiological recording in biomedical and physiological research and clinical practice.
2. High degree of physiological recording accuracy with stable waveforms. Allows the ability to observe cardiovascular activity on-screen.
3. Physiological recording is non-invasive, and thus comfortable for the individual, with reliable signal quality and recording artifact reduction, assuming it is utilized appropriately.
4. Multiple physiological signals (ECG, EEG, EMG, respiration and blood pressure) can be recorded and analyzed, providing flexibility in the design of a physiological experiment.
5. Highly versatile and excellent system for educational and research purposes, providing clear physiological tracings and insights into human physiology.
6. BIOPAC Applications
7. The BIOPAC system is widely used for many purposes in biomedical engineering, clinical physiology, neuroscience, psychology and medical education.
8. Within research into the cardiovascular system the BIOPAC system is often used forECG recordings, heart rate monitoring, blood pressure determination and the analysis of heart rate variability (HRV).
9. The recording of electroencephalograph (EEG) signals and analysis of neural brain activity is also enabled by the BIOPAC system for those studies involving the nervous system. Recording muscle activity using electromyography is useful in movement and neuromuscular research.

10. Studies investigating physiological response to stress utilize the BIOPAC system because the stress response directly affects cardiovascular and autonomic nervous system activity.
11. Academic institutions frequently use the BIOPAC system for student teaching purposes with physiological principles, instrumentation and the collection and interpretation of physiological data being core subjects.

The range of applications and the reliable nature of the BIOPAC system make it invaluable for the research community.

3.7ECG Recording procedure

ECG recordings were made in controlled laboratory conditions to achieve optimal recording conditions and physiological reliability with minimal recording artifact.

Subjects were asked to sit still in a comfortable position for the entire recording session and to remain motionless unless necessary, and allowed to reach a steady state before each recording commenced. Surface ECG electrodes were applied according to a modified Lead II configuration and ECG recordings were collected via the BIOPAC MP36 ECG-1 channel. ECGs of approximate length of 25 to 30 seconds were made on resting individuals.

Throughout the whole recording, data was observed on screen to monitor the tracing and heart rate. Care was taken with the tracing quality and clarity of the RR-intervals, as accurate analysis of HRV data relies on clear and stable tracings. Recording on examination day could not take place due to technical issues with the BIOPAC system during the recording period so examination stress response would need to be considered as literature-based evidence as to that found. This demonstration still allowed for practical ECG recording and HRV analysis to be obtained.

3.8 Kubios HRV Software

The analysis software utilized in the study was Kubios HRV Scientific Lite version 4.3.0, the gold standard in autonomic and cardiovascular variability analysis. Kubios is an internationally recognized software used to derive and interpret the parameters describing HRV. RR-intervals are derived from the ECG trace and these may be corrected for artifacts through advanced preprocessing methods to obtain the most accurate measurements possible. Parameters of time-domain and frequency-domain HRV analysis were derived as standard. Time-domain analysis variables included mean heart rate, minimum and maximum heart rate, standard deviation of the time between RR-intervals and RMSSD, while frequency-domain analysis was comprised of low-frequency power (LF), high-frequency power (HF) and the ratio LF/HF. RR-interval plot, histogram and Poincar plot outputs were produced graphically and provided insight into the pattern of autonomic activity.

Benefits of using Kubios HRV software include automated RR-interval identification and correction, improving the accuracy of cardiovascular variability measurements. The ability of this software to correctly interpret and present cardiovascular variability data has thus proven invaluable in the present investigation.

3.9 HRV parameters

Various heart rate variability parameters were used to investigate the dynamic changes within cardiovascular activity and the autonomic nervous system response to various stimuli. The average, minimum and maximum heart rates were derived as simple indications of cardiovascular activity.

Standard deviation of heart rate gives an indication of fluctuation between heart beats. The parameters SDNN and RMSSD are time-domain measures of HRV and SDNN gives an indication of total autonomic variability and parasympathetic influence while RMSSD shows variation between successive beats as a measure of predominantly parasympathetic activity.

LF and HF are frequency-domain parameters which describe fluctuations of varying frequencies, LF is a measure of both parasympathetic and sympathetic activity while HF is solely an indicator of parasympathetic activity. The LF/HF ratio was considered to indicate a measure of the relationship between sympathetic and parasympathetic activation. All these parameters were considered to provide an overall picture of cardiovascular activity and the autonomic nervous system.

3.10 Statistical analysis

Data was organized and interpreted in Kubios HRV software, in combination with Microsoft Excel where relevant and then analyzed using the methods described above. Heart rates and RR-intervals measured and derived in Kubios HRV software and HRV parameters derived from it were recorded and graphically represented. Descriptive statistics were then applied to measures such as standard deviation of heart rate, RMSSD, SDNN etc.

These results are presented in graphical form for comparison. Comparison of expected responses to examination stress was discussed in comparison with the findings presented in previous literature as an interpretation of results obtained from examination period recordings could not be produced. In summary statistical analysis aided objective determination of physiological responses of the college student during academic stress.

CHAPTER 4

OBSERVATIONS and RESULTS

4.1 Participant Screening

Participant screening was conducted prior to collecting physiological data, in order to ensure the suitability of the subjects for cardiovascular and autonomic nervous system analysis. This process was based on the Perceived Stress Scale (PSS-10) and a general health questionnaire. These measurements were essential to verify that the participants selected were healthy with low baseline stress levels and free of conditions that could potentially affect the results obtained from ECG and HRV measurements.

The Perceived Stress Scale (PSS-10) is a validated psychological measurement used to assess levels of perceived stress within an individual. It consists of 10 questions relating to a participant's feelings of emotional stress, how they cope with these feelings and their emotional reaction to everyday situations. Each question has a defined response scale and the total score provides an estimation of the participant's perceived stress.

The results of the PSS-10 screening in the present study, and consequently for inclusion in the final physiological analysis, were that only participants with low stress levels were included in this study. Those participants who scored less than or equal to 13 on the PSS-10 scale were included in the study as the objective was to gather normal baseline physiological measures with a relatively stable autonomic nervous system functioning.

A general health questionnaire was used to measure participants' medical history, cardiovascular condition, medication intake and presence of chronic illness, thereby excluding subjects with known medical conditions that would affect heart rate variability and other cardiovascular parameters.

Out of eleven college students who were part of the initial screening process, one participant was excluded due to a cardiac-related health condition, thus not providing a reliable baseline ECG and HRV measure. Thus a sample size of ten healthy subjects were used in the present study.

Overall, the screening of the participants ensured high levels of scientific reliability in the present study by minimizing confounding effects of chronic health conditions or abnormally high stress levels and creating physiological similarity among participants.

4.2 Normal Day ECG Results

ECG measurements were taken from all eligible participants under normal resting condition by using the BIOPAC MP36 physiological acquisition system. ECG recordings were acquired in a resting lab setting with a relatively calm and quiet environment in order to prevent any movement artifact or disturbance that could impact upon these cardiovascular measurements.

Before recordings of ECG were made the subjects were asked to relax and to try and limit any unwanted movement during recording. Standardised leads placements for each recording session were utilized to ensure consistent waveform interpretation, and to assist in correct detection of RR-interval using the modified Lead II configuration.

Each ECG session lasts from 25-30 seconds, during which the tracing was monitored using BIOPAC in order to confirm that the recordings were stable and clear. The presence of P waves, QRS complexes and T waves was identified indicating valid recording of cardiac activity.

A normal sinus rhythm pattern was exhibited by all the participants within their recordings and fluctuations in the heart rate was within a physiologically acceptable resting range. The minimum and maximum heart rate values were extracted from the recordings to be used with Kubios HRV Scientific Lite 4.3.0 to measure subsequent HRV parameters.

The ECG tracings recorded from each individual during a normal day give us a measure of baseline cardiac activity and autonomic nervous system functioning within each healthy college student. Variations in the heart rate at resting condition reflects inherent physiological variations based on the balance within the autonomic nervous system, influencing the cardiac response to changes around the individual.

BIOPAC successfully acquired ECG data suitable for use in RR-interval extraction and the measurement of various other HRV parameters. There were very minimal artifacts present within the traces due to a controlled stable physical position of the subject during acquisition of the traces.

In general the normal day ECGs will provide valuable insight for the comparison with ECGs recorded on stress or anxious days, by representing the normal cardiovascular variations within healthy human beings.

4.3 Analysis Results for HRV

Using the RR interval values extracted from the ECG recording through Kubios HRV Scientific Lite 4.3.0 software, HRV analysis was performed. HRV analysis is among the most commonly studied, important non-invasive methods for assessing autonomic activity and cardiovascular regulation.

The ECG recordings from each participant were input into Kubios software where they were processed for RR interval measurement and other signal manipulations. To improve the reliability and quality of the recordings and minimize distortion resulting from minor errors in the signal recording, artifacts correction procedures within the software were used.
Analysis with the time-domain HRV parameters from the recordings showed mean, minimum, maximum heart rate values, SD and RMSSD and SDNN.

Mean, min, and max heart rates were within a range to support a normal resting state. The obtained SD, SDNN, RMSSD, min, and max HR values supported normal variation in cardiovascular system functions associated with the autonomic nervous system.

SDNN is a measure of standard deviation for RR interval, while RMSSD is the root mean square successive difference, representing the short-term fluctuations of R-R intervals and mostly reflect the parasympathetic activity. These values were found to indicate a relatively balanced state of autonomic nervous system under a relaxed state and did not seem to represent signs of cardiovascular abnormality, sympathetically dominant state or anything else outside normal physiology.
Various graphs including RR interval graphs, HRV trend graph, histograms and Poincar graphs were generated by the software. The graphs visualized and explained the trends in heart rate variability and autonomic nervous system.

The plots suggested that during the rest condition, the cardiovascular systems of the participants exhibited regular and stable states. The overall physiological results seem to indicate reliability in obtaining ECG signals and applying HRV analysis. This indicates that findings are congruent with published data which also reported normal resting state of the cardiovascular system in healthy young adults.

Parameters analyzed:
Mean Heart Rate (HR)
Minimum Heart Rate
Maximum Heart Rate
Standard Deviation (SD)
SDNN
RMSSD
RR Interval Variability

Observation and Result Analysis (Normal Day):

Table 4.3.1: Extracted HRV Parameters Including PNS Index, SNS Index, SDNN, and RMSSD

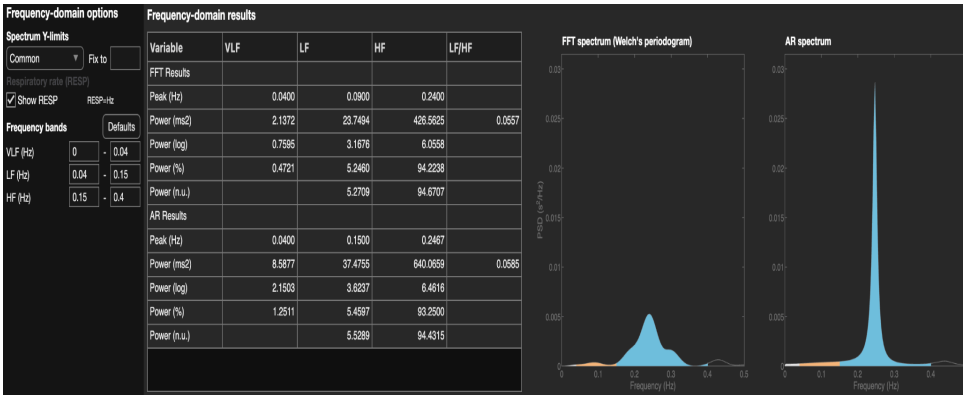
HRV PARAMETERS OF 10 PARTICIPANTS

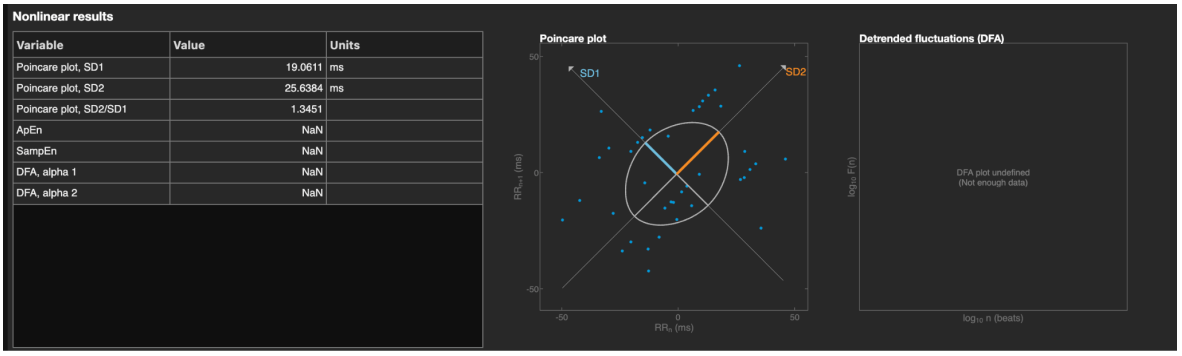
Participant	Min HR (BPM)	Max HR (BPM)	PNS Index	SNS Index	SDNN (ms)	RMSSD (ms)
P1	65.97	97.73	-0.98	2.31	24.36	30.50
P2	57.85	94.28	2.19	1.82	19.45	23.41
P3	60.20	97.78	-1.51	3.73	21.49	26.08
P4	58.24	95.26	-1.40	2.95	21.12	27.22
P5	57.07	96.55	-1.43	2.53	20.67	24.73
P6	56.63	94.56	-1.47	2.88	20.59	25.03
P7	57.42	96.84	-1.54	3.07	25.47	26.03
P8	57.82	97.43	-1.52	2.33	20.90	25.24
P9	57.71	92.52	-1.49	3.27	24.18	28.06
P10	64.52	101.52	-1.45	2.14	23.23	24.93

Note: Min HR = Minimum Heart Rate, Max HR = Maximum Heart Rate, PNS = Parasympathetic Nervous System,
SNS = Sympathetic Nervous System, SDNN = Standard Deviation of NN intervals, RMSSD = Root Mean Square of Successive Differences

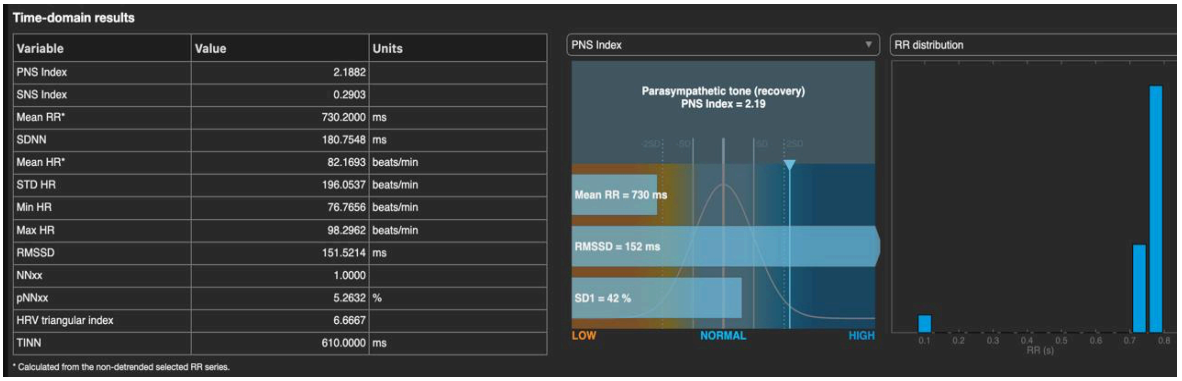
Kubios Software data analysis :

Participant 1 Data :



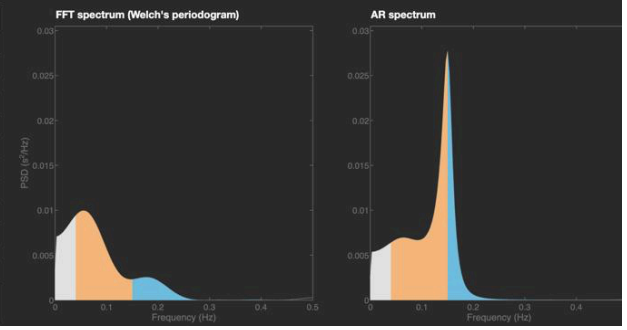


Participant 2 Data :



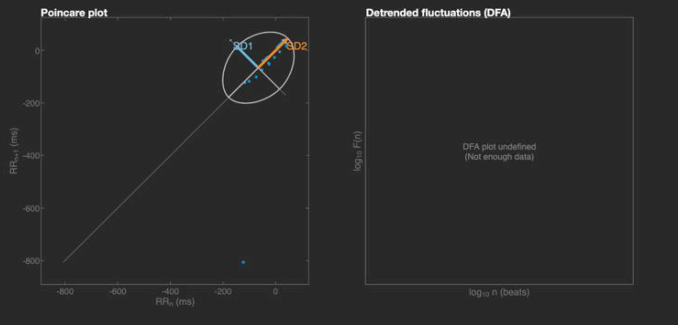
Frequency-domain results

Variable	VLF	LF	HF	LF/HF
FFT Results				
Peak (Hz)	0.0400	0.0567	0.1767	
Power (ms2)	313.7931	689.6482	184.3926	3.7401
Power (log)	5.7487	6.5362	5.2171	
Power (%)	26.4166	58.0579	15.5231	
Power (n.u.)		78.9009	21.0959	
AR Results				
Peak (Hz)	0.0400	0.1500	0.1500	
Power (ms2)	224.4734	1.0556e+03	391.0180	2.6995
Power (log)	5.4138	6.9618	5.9688	
Power (%)	13.4329	63.1670	23.3993	
Power (n.u.)		72.9688	27.0303	



Nonlinear results

Variable	Value	Units
Poincare plot, SD1	114.3699	ms
Poincare plot, SD2	158.7912	ms
Poincare plot, SD2/SD1	1.3884	
ApEn	NaN	
SampEn	NaN	
DFA, alpha 1	NaN	
DFA, alpha 2	NaN	

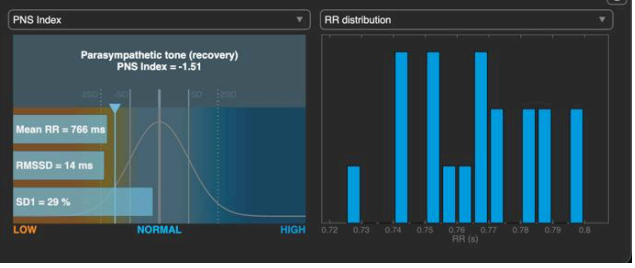


Participant 3 Data :

Time-domain results

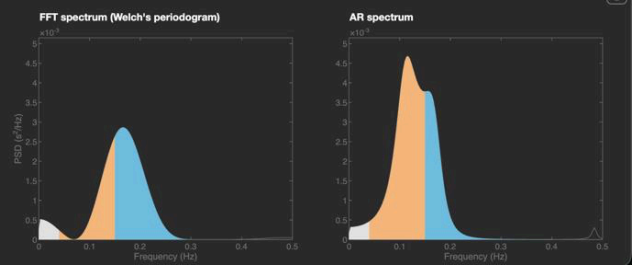
Variable	Value	Units
PNS Index	-1.5141	
SNS Index	3.5701	
Mean RR*	766.0500	ms
SDNN	19.9001	ms
Mean HR*	78.3239	beats/min
STD HR	2.0468	beats/min
Min HR	76.5306	beats/min
Max HR	80.7537	beats/min
RMSSD	13.8404	ms
NNxx	0	

* Calculated from the non-detrended selected RR series.



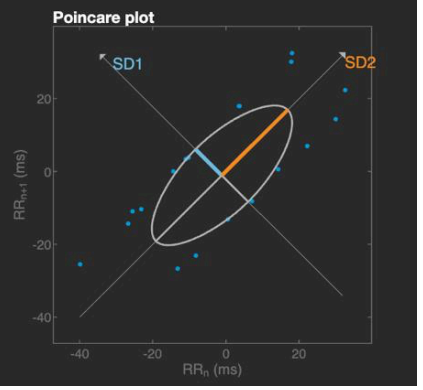
Frequency-domain results

Variable	VLF	LF	HF	LF/HF
FFT Results				
Peak (Hz)	0.0033	0.1500	0.1667	
Power (ms2)	16.2775	87.1339	187.0674	0.4658
Power (log)	2.7898	4.4674	5.2315	
Power (%)	5.6036	29.9961	64.3985	
Power (n.u.)		31.7768	68.2214	
AR Results				
Peak (Hz)	0.0400	0.1167	0.1567	
Power (ms2)	14.1903	293.7040	137.1008	2.1422
Power (log)	2.6526	5.6826	4.9207	
Power (%)	3.1888	66.0010	30.8092	

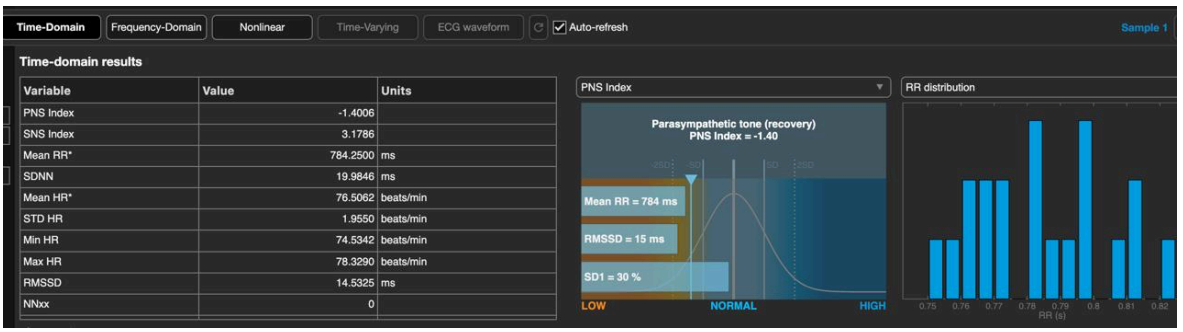


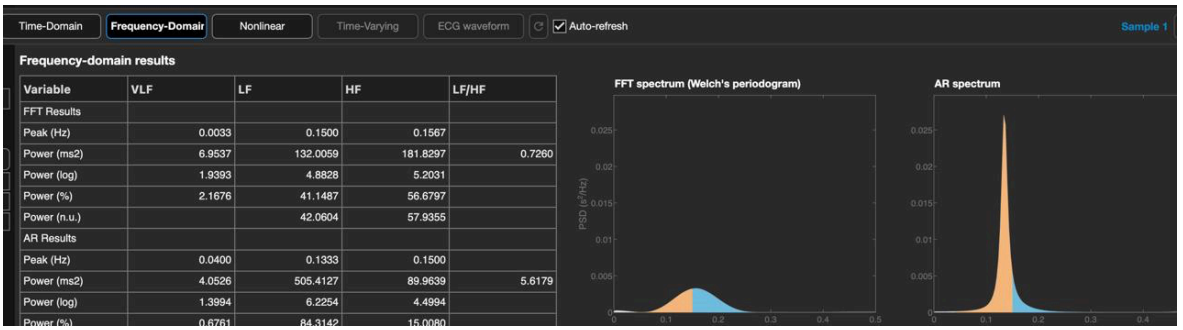
Nonlinear results

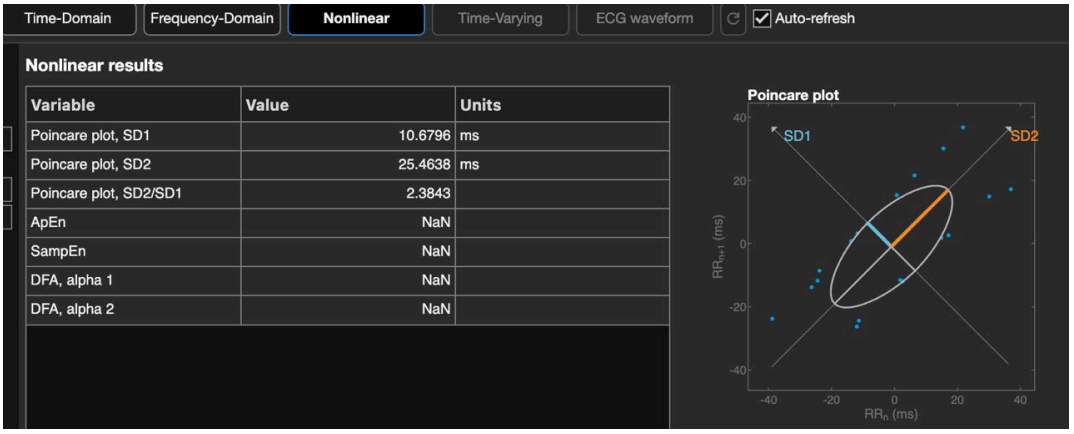
Variable	Value	Units
Poincare plot, SD1	10.1699	ms
Poincare plot, SD2	25.4725	ms
Poincare plot, SD2/SD1	2.5047	
ApEn	NaN	
SampEn	NaN	
DFA, alpha 1	NaN	
DFA, alpha 2	NaN	



Participant 4 data :

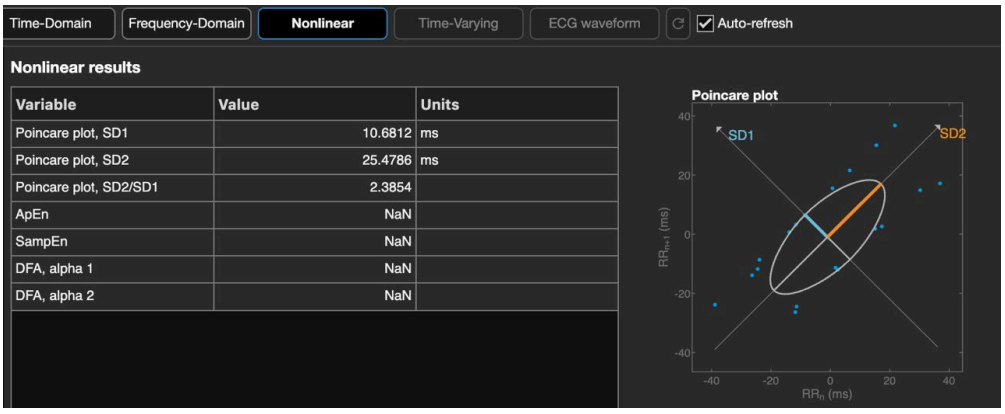




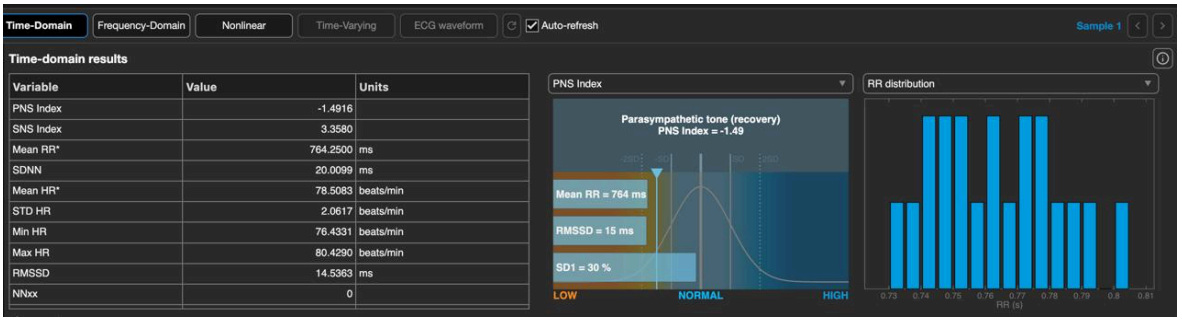


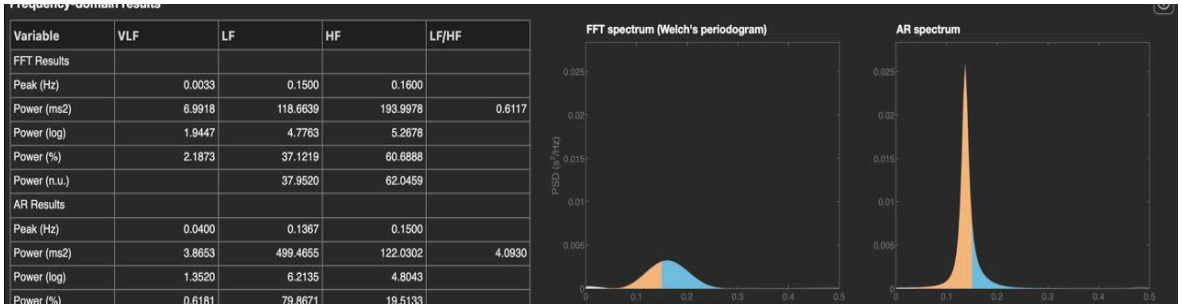
Participant 5 Data :

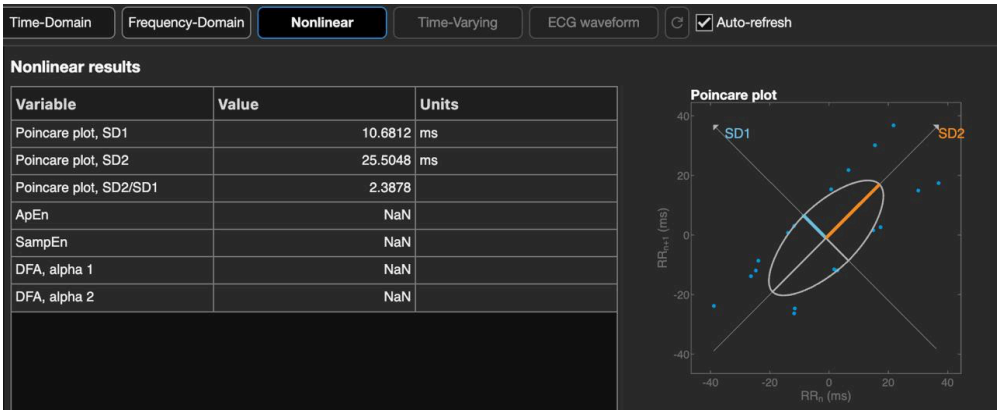




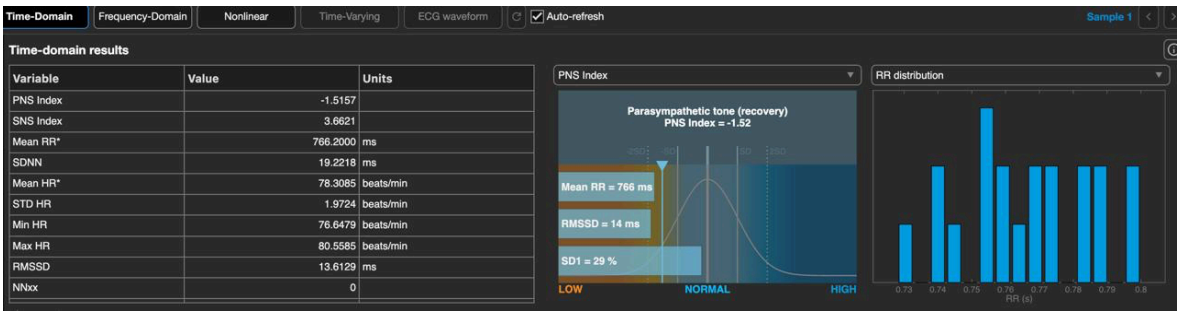
Participant 6 Data :

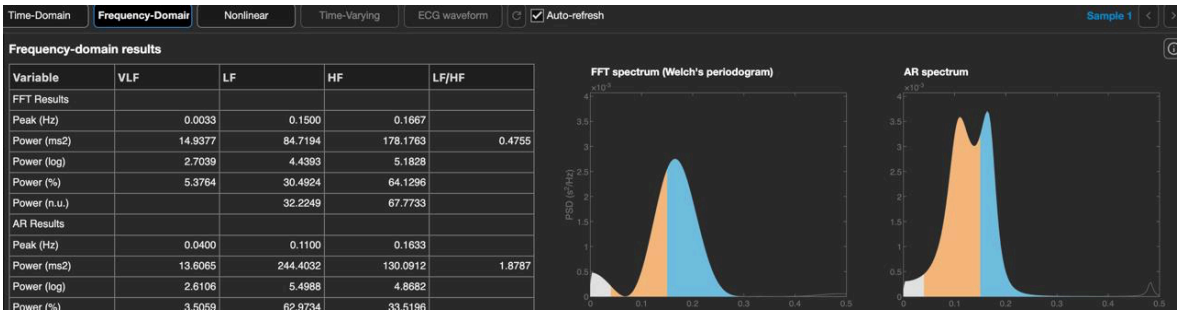


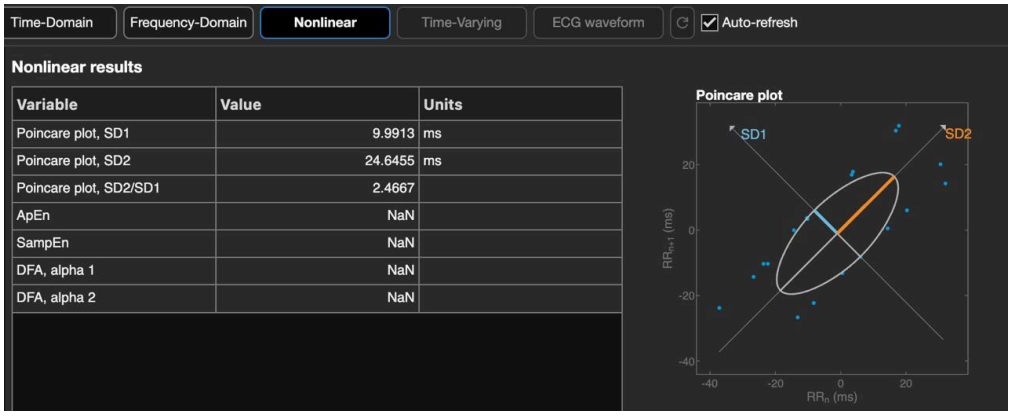




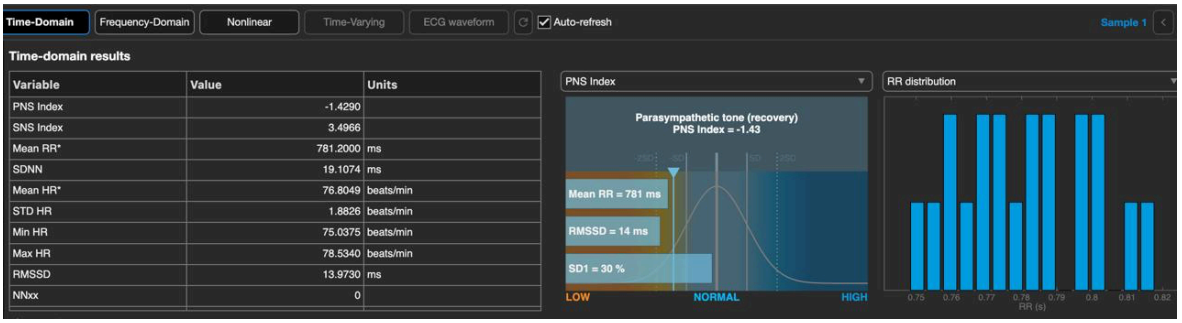
Participant 7 Data :

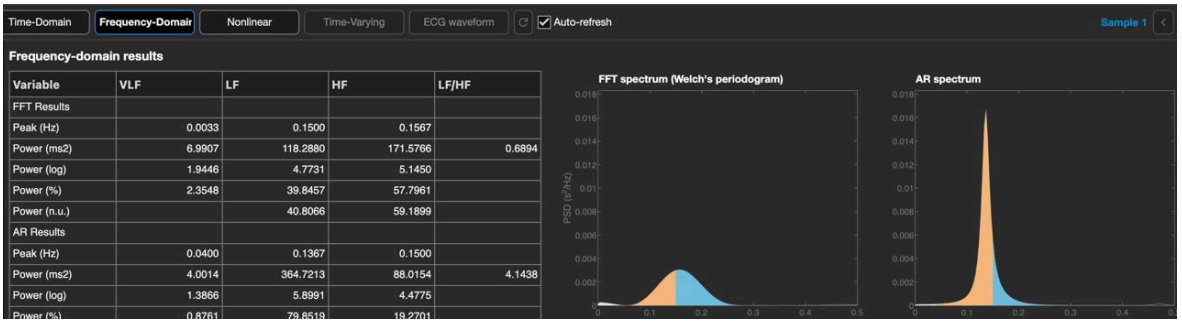




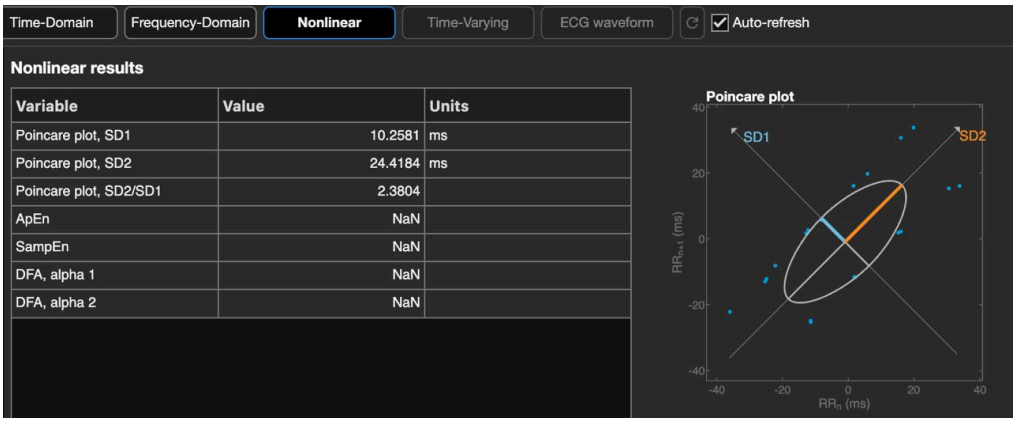


Participant 8 Data :





Variable	VLF	LF	HF	LF/HF
FFT Results				
Peak (Hz)	0.0033	0.1500	0.1567	
Power (ms2)	6.9907	118.2880	171.5766	0.6894
Power (log)	1.9446	4.7731	5.1450	
Power (%)	2.3548	39.8457	57.7961	
Power (n.u.)		40.8066	59.1899	
AR Results				
Peak (Hz)	0.0400	0.1367	0.1500	
Power (ms2)	4.0014	364.7213	88.0154	4.1438
Power (log)	1.3866	5.8991	4.4775	
Power (%)	0.8761	79.8519	19.2701	

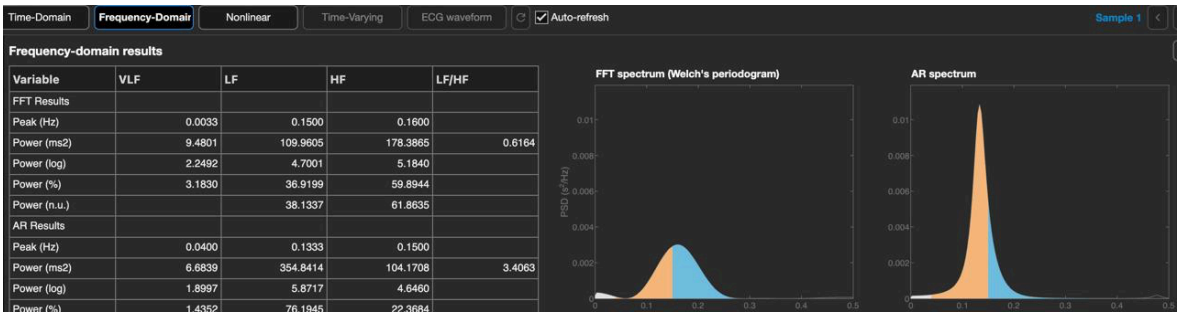


Variable	Value	Units
Poincare plot, SD1	10.2581	ms
Poincare plot, SD2	24.4184	ms
Poincare plot, SD2/SD1	2.3804	
ApEn	NaN	
SampEn	NaN	
DFA, alpha 1	NaN	
DFA, alpha 2	NaN	

Participant 9 data :



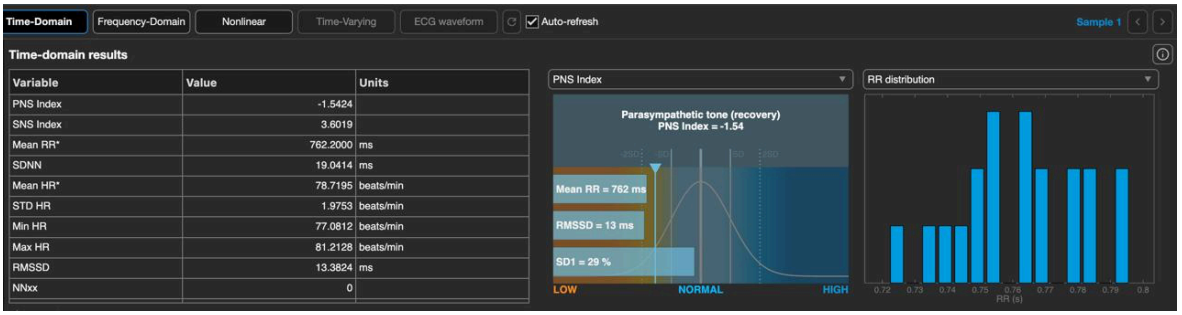
Variable	Value	Units
PNS Index	-1.4662	
SNS Index	3.1964	
Mean RR*	772.4500	ms
SDNN	19.7139	ms
Mean HR*	77.6749	beats/min
STD HR	1.9891	beats/min
Min HR	75.8534	beats/min
Max HR	79.6813	beats/min
RMSSD	14.1923	ms
NNxx	0	

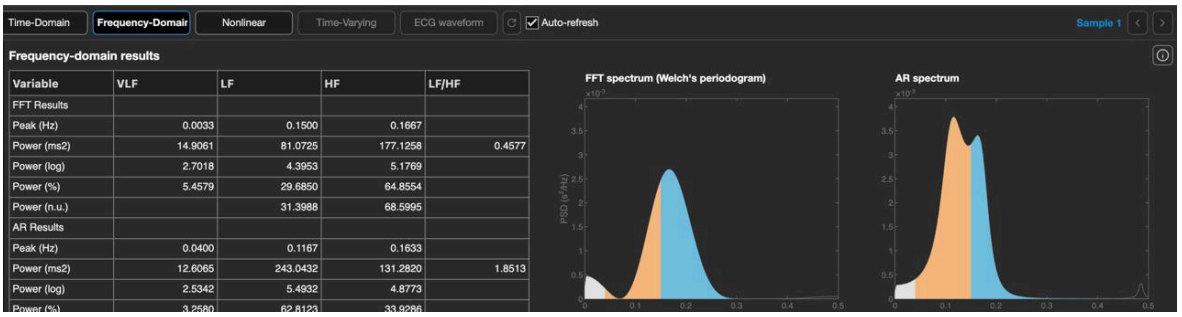


Variable	VLF	LF	HF	LF/HF
FFT Results				
Peak (Hz)	0.0033	0.1500	0.1600	
Power (ms2)	9.4801	109.9605	178.3865	0.6164
Power (log)	2.2492	4.7001	5.1840	
Power (%)	3.1830	36.9199	59.8944	
Power (n.u.)		38.1337	61.8635	
AR Results				
Peak (Hz)	0.0400	0.1333	0.1500	
Power (ms2)	6.6839	354.8414	104.1708	3.4063
Power (log)	1.8997	5.8717	4.6460	
Power (%)	1.4352	76.1945	22.3684	



Participant 10 data :





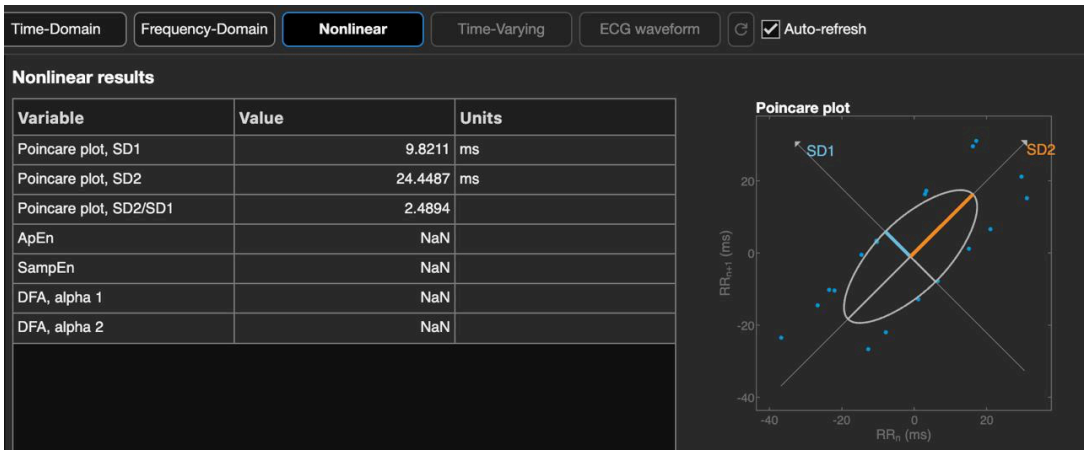


Figure 4.3.2 : Kubios HRV Frequency-Domain& poincare plot Analysis of Participant 1-10

Participant-wise Interpretation

Participant P1:
Participant P1 showed minimum and maximum heart rate values of 65.97 bpm and 97.73 bpm respectively. The SDNN value was 24.36 ms and RMSSD was 30.50 ms, both of which fall within acceptable normal physiological range. The PNS index (-0.98) and SNS index (2.31) indicate relatively balanced autonomic nervous system activity with mild sympathetic influence. Overall HRV findings suggest stable cardiovascular regulation and absence of significant physiological stress under resting conditions.

Participant P2:
Participant P2 demonstrated minimum and maximum heart rate values of 57.85 bpm and 94.28 bpm respectively. SDNN value was 19.45 ms, which is slightly lower than the ideal normal range, while RMSSD value was 23.41 ms indicating moderate parasympathetic activity. The elevated positive PNS index (2.19) together with relatively lower SNS index (1.82) suggests good parasympathetic regulation and relaxation response. Overall findings indicate near-normal autonomic balance with mild reduction in HRV adaptability.

Participant P3:
Participant P3 exhibited a minimum heart rate of 60.20 bpm and maximum heart rate of 97.78 bpm. SDNN and RMSSD values were 21.49 ms and 26.08 ms respectively, indicating satisfactory HRV and autonomic regulation. However, SNS index value of 3.73 suggests comparatively higher sympathetic nervous system activity. The negative PNS index (-1.51) indicates mild reduction in parasympathetic tone. These findings may represent mild physiological sympathetic dominance despite overall normal resting HRV.

Participant P4:
Participant P4 demonstrated minimum and maximum heart rate values of 58.24 bpm and 95.26 bpm respectively. SDNN value was 21.12 ms and RMSSD value was 27.22 ms, suggesting acceptable autonomic flexibility and parasympathetic activity. The SNS index (2.95) indicates moderate sympathetic activation, while the PNS index (-1.40) suggests slightly reduced parasympathetic influence. Overall interpretation indicates relatively stable cardiovascular autonomic function under resting conditions.

Participant P5:
Participant P5 showed minimum heart rate of 57.07 bpm and maximum heart rate of 96.55 bpm. SDNN value was 20.67 ms and RMSSD was 24.73 ms, both indicating borderline-normal HRV status. The SNS index value of 2.53 suggests moderate sympathetic activity, whereas the PNS index (-1.43) indicates mildly decreased parasympathetic activity. Overall findings suggest acceptable autonomic balance without evidence of severe physiological stress.

Participant P6:
Participant P6 demonstrated minimum and maximum heart rate values of 56.63 bpm and 94.56 bpm respectively. SDNN value was 20.59 ms and RMSSD value was 25.03 ms, which indicate maintained HRV and parasympathetic activity. The SNS index (2.88) indicates moderate sympathetic influence, while the PNS index (-1.47) suggests mild reduction in parasympathetic tone. Overall cardiovascular regulation appears physiologically stable under resting conditions.

Participant P7:
Participant P7 exhibited a minimum heart rate of 57.42 bpm and maximum heart rate of 96.84 bpm. SDNN value was 25.47 ms and RMSSD value was 26.03 ms, indicating good heart rate variability and healthy autonomic adaptability. The SNS index value of 3.07 suggests increased sympathetic nervous system activity, whereas the PNS index (-1.54) indicates reduced parasympathetic dominance. Despite mild sympathetic predominance, HRV findings remain within acceptable physiological range.

Participant P8:
Participant P8 showed minimum and maximum heart rate values of 57.82 bpm and 97.43 bpm respectively. SDNN and RMSSD values were 20.90 ms and 25.24 ms respectively, indicating maintained autonomic regulation. The SNS index (2.33) suggests moderate sympathetic activity and the PNS index (-1.52) indicates mild reduction in parasympathetic influence. Overall findings are consistent with stable resting cardiovascular function.

Participant P9:
Participant P9 demonstrated a minimum heart rate of 57.71 bpm and maximum heart rate of 92.52 bpm. SDNN value was 24.18 ms and RMSSD value was 28.06 ms, indicating satisfactory HRV and good parasympathetic activity. However, the SNS index value of 3.27 suggests increased sympathetic activation. The PNS index (-1.49) reflects mild reduction in parasympathetic tone. Overall interpretation indicates normal autonomic regulation with mild sympathetic predominance.

Participant P10:
Participant P10 exhibited minimum and maximum heart rate values of 64.52 bpm and 101.52 bpm respectively. SDNN value was 23.23 ms and RMSSD value was 24.93 ms, indicating acceptable HRV status and autonomic balance. The SNS index (2.14) suggests moderate sympathetic activity, while the PNS index (-1.45) indicates mild reduction in parasympathetic regulation. Overall findings demonstrate relatively stable cardiovascular and autonomic function under resting conditions.

Overall Interpretation of the Study:
The autonomic nervous system regulation under resting conditions was assessed by HRV analysis, and was found to be stable on average across the participants. The majority of participants had SDNN values of more than 20ms and RMSSD values of around or greater than 25ms which represent acceptable heart rate variability and cardiovascular adaptability. A few participants showed signs of mild sympathetic predominance (higher SNS index) and during their resting state/quiescent state, none of them showed signs of severe autonomic imbalance and/or significant physiological stress.

ECG and HRV analysis showed the present results are consistent with the value of this objective physiological methods of assessment of the autonomic nervous system and cardiovascular evaluation in connection with functional state of stress.

CHAPTER 5

DISCUSSION

The present study was done to assess the HR and HRV among healthy college students, recorded by ECG parameters with the BIOPAC MP36 physiological data acquisition system and analyzed using the Kubios HRV Scientific Lite 4.3.0 software. This study was mainly concerned with measuring autonomic nervous system activity and cardiovascular variability at rest. The decision to use ECG-based HRV analysis was made due to its non-invasive and objective assessment of autonomic nervous system control and cardiovascular activity that is related to stress. HRV is regarded as one of the most important indicators of autonomic nervous system balance. RR interval changes are related to the interplay between the sympathetic and parasympathetic nervous system. A healthy cardiovascular system has a flexible autonomic control system, with normal fluctuations in the heart rate. HRV is often lower in stressed, fatigued, anxious individuals as well as in those with autonomic imbalance and cardiovascular dysfunction, and higher HRV levels are correlated with greater physiological adaptability and recovery capacity.

Successful acquisition of ECG recordings in the present study was realized from ten healthy subjects at rest. ECG signals were analyzed by Kubios HRV software for the calculation of time domain, frequency domain and nonlinear HRV parameters. Assessment of autonomic nervous system activity and cardiovascular regulation was done through parameters such as minimum heart rate, maximum heart rate, SDNN, RMSSD, PNS index, SNS index, RR interval variability and Poincaré plot analysis.

The results found indicated that the participants had a relatively stable cardiovascular and autonomic function, at rest. The majority of participants had SDNN values greater than 20ms and RMSSD values of 25ms or greater, reflecting good autonomic functioning. These results indicate a good parasympathetic function and appropriate cardiovascular adaptability during rest. The lowest and highest heart rate readings were also within physiological limits for healthy young adults.

The SDNN parameter represents total HRV and autonomic adaptability. In the current study, most participants exhibited SDNN values between about 20 and 25 ms, which reflects good autonomic flexibility. A decrease in SDNN is frequently observed with autonomic imbalance and a decrease in cardiovascular adaptability due to stress. Based on the results therefore, the participants were not severely stressed physiologically during the resting measures.

Likewise, the values of RMSSD observed in the study were within the acceptable range, which reflects parasympathetic nervous system activity and short-term heart rate variability. The relationship between RMSSD and vagal activity/relaxation responses is strong. An RMSSD value is generally considered healthier the higher it is; a lower RMSSD value may be a sign of stress related sympathetic dominance. The above results, therefore, confirm the relatively stable activity of the parasympathetic system in the participants during normal rest.

Also, the autonomic interpretation was greatly aided by the PNS and SNS indices derived from Kubios HRV analysis. The majority of the participants had mild sympathetic predominance and relatively preserved parasympathetic function. Some participants had slightly higher SNS index values, but HRV results did not signify an excessive autonomic imbalance and/or physiological stress. Mild sympathetic activation may be physiological, as a result of diurnal variations in the body, ambient temperature, expectations during recording and emotional states. But the autonomic balance did not change in a physiologically abnormal way in the subjects as a whole.

Further interpretation of cardiovascular variability was obtained from the nonlinear HRV analysis carried out with Kubios software. Poincaré plot analysis revealed that the RR interval variability was organized in an elliptical distribution for participants. Short term HRV component was indicated by SD1 while long term component by SD2, which were derived from nonlinear analysis. Resting cardiovascular rhythm dynamics (autonomic regulation) were organized as evidenced by the presence of relatively stable elliptical patterns.The frequency-domain analysis was also useful for information about autonomic modulation. Typical parameters for the assessment of sympathetic/parasympathetic balance are LF power, HF power, and LF/HF ratio. HF components tend to be linked with parasympathetic activity and LF components may be linked with both sympathetic and parasympathetic activity. During the stress, the LF/HF ratio is also often correlated with sympathetic dominance. Frequency-domain results were not physiologically outside the normal range in the present study, suggesting the maintenance of autonomic stability during rest recordings.

This study proved the usefulness of ECG and HRV analysis in physiological and biomedical studies. ECG recordings from BIOPAC yielded high quality, good cardiac electrical activity signals and RR interval extraction. The software package Kubios HRV Scientific Lite 4.3.0 facilitated efficient processing of the RR interval data obtained from the ECG and detailed autonomic analysis by means of time domain, frequency domain and nonlinear HRV methods. Thus, the use of BIOPAC in combination with the Kubios software was found to be effective for assessment of cardiovascular variability and evaluation of the autonomic nervous system.

It has been reported by several previous studies that academic stress has a significant impact on the activity of the autonomic nervous system and cardiovascular regulation in the students. Increased sympathetic nervous system activity, elevated HR, elevated BP, decreased HRV and decreased parasympathetic activity have all been linked to examination stress. Psychological stress activates the hypothalamic-pituitary-adrenal axis and sympathetic nervous system, resulting in increased secretion of stress hormones such as adrenaline and cortisol. These are physiological reactions that are a component of the "fight-or-flight" mechanism of stress adaptation.ECG and HRV analysis during examination periods has been shown to significantly decrease HRV parameters (including SDNN and RMSSD) and increase the LF/HF ratio and heart rate in research studies. Stress-related changes in HRV indicate decreased autonomic flexibility and increased sympathetic dominance. As a result, HRV analysis has been the subject of several studies as an objective marker of the evaluation of stress and autonomic nervous system monitoring.

During the present study, the purpose of the ECG recordings was originally to compare the physiological responses of the participants at rest and during stress. Technical issues with the BIOPAC acquisition system during the period of the examination prevented experimental examination-day recordings from being completed, however. Thus, within the scope of the present investigation direct physiological comparison between resting and examination condition was not possible.

Thus, the physiological response that are reviewed in this study are only those previously published in scientific literature and are those that are expected when being stressed, but not those obtained from experiments. Based on the literature, examination stress should be known to cause the following changes: increased HR, increased sympathetic activity, decreased parasympathetic activity, decreased HRV parameters, and increased ratio LF/HF. Such physiological changes have been reported in the majority of stress related cardiovascular research studies conducted with students and academic stress situations.
Resting HRV profiles of healthy college students were successfully established in the present study, and the effectiveness of HRV analysis using ECG for autonomic assessment was demonstrated, although it was impossible to obtain such data experimentally on the day of the examination. The obtained resting physiological data serve as useful reference information for the cardiovascular variations and autonomic balance of healthy subjects in non-stress conditions.

The present study also highlights the importance of HRV as a non-invasive physiological biomarker for stress research. HRV analysis can offer objective information about autonomic nervous system (ANS) functioning, emotional regulation, cardiovascular adaptability, and physiological stress response. HRV provides measurable physiological data on autonomic activity and cardiovascular regulation that is not captured through questionnaires of subjective stress. Thus, the use of psychological assessment measures in combination with ECG and HRV analysis could enhance the stress assessment in future studies.

There are some limitations to this study. The number of participants in the sample was small (only 10). In addition, short duration resting measurements were made by ECG. Also, due to the technical problems with the BIOPAC system, physiological recordings on the day of the examination were not completed. These constraints limited the ability to conduct experimental stress comparisons and statistical analyses.

Larger number of subjects, longer duration of ECG recording, repeated measurements and successful real time examination period recordings could therefore be included in future studies for a fuller assessment of the level of stress. Further research can take into account more physiological parameters, including blood pressure, respiration rate, galvanic skin response, cortisol level and continuous monitoring, in order to gain a wider picture of the academic stress physiology.

More sophisticated HRV methods such as machine learning stress classification, monitoring using wearable biosensors, and long-term ambulatory ECG analysis could enhance the accuracy of detecting stress and physiological outputs. Future studies could also explore gender differences, sleep quality, anxiety and learning outcomes for students with respect to HRV and autonomic regulation.

In general, the present study showed that ECG and HRV analysis could be used as useful and reliable tools in physiological monitoring and assessment of autonomic nervous system in healthy college student's population. The results of the HRV at rest showed a relatively stable cardiovascular variability and a good autonomic regulation at rest. Mention of recordings obtained during direct examination was impossible to complete in an experimental fashion, but literature-supported expectations of autonomic response to stress, and cardiovascular response, indicate that academic stress may have a significant effect on autonomic balance and cardiovascular function. The results of this study thus serve as a beneficial basis for further research on physiological stress assessment using ECG and HRV analysis in schools.

CHAPTER 6

SUMMARY AND CONCLUSION

The present study was carried out to assess HRV and cardiovascular responses among healthy college students with the technique of electrocardiography and HRV analysis. This study aimed to investigate the autonomic nervous system (ANS) activity under resting physiological conditions using the BIOPAC MP36 physiological recording system and HRV analysis software Kubios HRV Scientific Lite 4.3.0.

Stress is an important physiological and psychological factor that can affect the regulation of the autonomic nervous system and cardiovascular function. The stress in school has been defined as one of the most important psychological and physical stresses among students, especially during exams. There have been some previous reports that examination stress can cause substantial changes to HR, BP, autonomic balance and HRV. Hence, the current research was done to assess healthy college students' HRV parameters and to study the applicability of ECG-based HRV analysis as a non-invasive physiological tool for stress assessment.
After screening via health questionnaires and Perceived Stress Scale (PSS-10), 10 healthy college students participated in the study. To reduce physiological variability that was not related to stress, participants who had cardiovascular disease, chronic illness, or who were taking medication that affected their heart rate were not included for this study. Prior to the collection of data, ethical considerations and informed consent procedures were observed.

The BIOPAC MP36 system (ECG-1 channel and modified Lead II electrode placement) was successfully used to record ECGs under normal resting conditions. The data from the ECG waveforms recorded by the participants had a good signal quality and RR interval data could be extracted for HRV analysis. The time-domain, frequency-domain, and nonlinear HRV analysis was performed using the ECG recordings analysed with Kubios HRV Scientific Lite 4.3.0 software. Several important parameters related to HRV were observed in the study, such as minimum heart rate, maximum heart rate, SDNN, RMSSD, PNS index, SNS index, LF power, HF power, LF/HF ratio, RR interval variability and nonlinear Poincaré plot analysis. These are the commonly reported parameters of autonomic nervous system activity and cardiovascular adaptability.

The results under resting conditions showed fairly stable cardiovascular control and good autonomic balance in the subjects. Most participants exhibited SDNN values greater than 20ms, and RMSSD values of approximately or greater than 25ms indicating sufficient PNS tone and heart rate variability. Maximum heart rate recordings were also within the normal limits of healthy young adults. Nonlinear HRV analysis also revealed organized HRV and stable autonomous function patterns, as did the Poincare plot evaluation.

According to the PNS and SNS indices derived from Kubio's analysis there was relatively balanced autonomic nervous system regulation, with a mild sympathetic predominance in some participants. But no participant exhibited large autonomic imbalance or abnormal physiological stress during the resting recording. In general, HRV measurements were within normal limits for the cardiovascular adaptability and autonomic regulation at rest.

The present study further showed the applicability of ECG and HRV analysis from the perspective of physiological and biomedical study. BIOPAC MP36 allowed reliable acquisition of ECG signals and Kubios HRV software offered complete autonomic analysis of HRV signals via advanced signal processing and HRV interpretation techniques. These systems were combined and shown to be successful for cardiovascular monitoring and physiological assessment without touching the patient.The original aim of the study was to compare normal resting conditions and examination-stress. But it was technically impossible to complete the examination-day ECG recordings as the BIOPAC system failed to function during data collection during the examination period. Thus, it was not possible to make a direct experimental comparison to normal and examination conditions within the present investigation.

Consequently, the physiological responses that have been discussed in this study have been gathered from previously published scientific literature and pre-established physiological responses to examination and not from experiments obtained. There is a large body of literature supporting that examination stress is linked to increased activity of the sympathetic nervous system, higher heart rate, decreased parasympathetic activity, decreased HRV parameters, and increased LF/HF ratio. Such physiological changes are regarded as autonomic imbalance and cardiovascular activation associated with stress.

Although there were no recordings of the examination day, the present study successfully set up healthy college students' baseline HRV profiles under rest state and showed that ECG and HRV analysis can be used to assess the autonomic nervous system. The acquired results confirm the value of HRV as an autonomous regulation and cardiovascular variability indicator.

The study also indicates the future opportunity of ECG and HRV analysis for monitoring stress in the student and other population. HRV is a physiological indicator that can be objective measures of autonomic nervous system functioning and can be important in the assessment of stress, mental health, cardiovascular monitoring, and prevention research.

The present study confirmed that HRV analysis of the ECG is a useful, reliable and non-invasive tool for monitoring CVV and ANS functions in healthy college students. Relatively stable autonomic regulation and cardiovascular adaptability were suggested by resting physiological measures used. Academic stress can potentially have an important effect on heart rate and autonomic nervous system imbalance as evidenced by physiological literature, even if examination-day recordings could not be experimentally completed. The study thus serves as a valuable basis for future research conducted in academic settings with the objective of real-time stress monitoring and sophisticated HRV analysis.

CHAPTER 7

FUTURE SCOPE

The results of the present study are helpful for future research on stress evaluation by ECG of college students. The study was able to test the patterns of HRV at rest in normal physiological conditions, but there are some possibilities that could improve and expand the research methodology in the future.

The next step in the research is to get actual ECG data during examination time, which will enable direct assessment of physiological stress responses of students. Comparative analysis of actual examination-stress conditions could be accomplished in future investigations to gain a clearer understanding of autonomic nervous system changes related to academic stress as we could not collect exam day data due to Biopac technical issue in lab.Such studies can give more precise physiologic data on cardiovascular responses to stress.

In future research, a larger sample size with participants from other academic disciplines, age groups and educational backgrounds could be included. A larger population would yield more statistical reliability and greater understanding of the variability among students in physiological stress. It is important to note that both men and women, with different educational backgrounds, are included in the study, which could help to reveal gender differences and differences across demographics in autonomic nervous system responses.One of the other future scope is the use of long duration and continuous ECG Monitoring system. An ambulatory monitoring system, such as wearable ECG devices and portable biosensors, will enable real-time monitoring of students during lectures, exams, presentations, and any other stressful academic situations. Continuous monitoring would allow for more detailed information relating to the dynamic cardiovascular and autonomic alterations during stress exposure.

In future research, advanced HRV analysis methods could be considered. HRV data could also be leveraged for the development of machine learning algorithms and artificial intelligence-based stress prediction systems that could enable automatic classification of stress and physiological risk assessment. Physiological monitoring systems that use artificial intelligence could potentially enhance the early identification of autonomic imbalance linked to stress and facilitate preventive health care applications.

Additional physiological and biochemical stress indicators like blood pressure, respiratory rate, galvanic skin response, oxygen saturation, cortisol secretion, salivary biomarkers, and electroencephalography (EEG) could also be included in future investigations. More complete assessment of the physiology and autonomic regulation of stress would be gained by using multi-parameter physiological assessment.

The longer durations of ECG recordings may yield even higher reliability of HRV and nonlinear analysis accuracy. Advances in the interpretation of HRV are limited because of short duration recordings of ECG, while long recordings better assess autonomic function and result in more stable measurements of cardiovascular variability.

Extended ECG acquisition protocols could then be used to provide more detailed physiological information from which new studies could be conducted.

The impact of sleep quality, anxiety, physical activity, academic outcome, emotional well-being and lifestyle on HRV patterns in school-aged children could also be assessed in future research. These studies could play an important role in the understanding of the link between mental health, autonomic regulation and cardiovascular function in academic settings.The cloud-based physiological monitoring systems and mobile health technologies could further enhance HRV data accessibility and real-time analysis. ECG systems for cell phones and wearable biosensors could make stress monitoring programs feasible on a large scale at schools and health care facilities.

In summary, with more participants and better physiological monitoring tools, real-time examination recordings, and some form of artificial intelligence systems, future studies could provide a much greater insight into academic stress physiology and autonomic nervous system regulation. This study is thus a valuable preliminary step to further extensive cardiovascular and stress-related physiological research with students.

LIMITATIONS OF THE STUDY

Several limitations were identified in the present study, which should be taken into account when drawing conclusions from this study. The main drawback was the limited number of participants (10). The larger number of participants may give more statistical reliability and more physiological interpretation.

One factor that limited the data was not being able to make an ECG recording during the actual examination period as a result of a technical set-up problem with the BIOPAC acquisition system. Therefore, it was not possible to conduct a direct experimental comparison between resting and examination-stress conditions within the scope of this study. Physiological responses discussed as part of the examination were therefore based on results obtained from literature, and not on experimental results.

The length of the ECG recording that was used in the study was relatively short, so some of the higher-order HRV analyses might not be suited and the interpretation of the autonomic activity over time may be limited. ECGs of longer duration can offer more comprehensive and accurate HRV measurements.

A focus of the study was the ECG and HRV parameters, and no other physiological stress parameters were recorded, such as blood pressure, cortisol levels, respiratory variability or psychological anxiety scales. More physiological measures may enhance the stress assessment.

Other factors like emotional state, environmental conditions, daily activities, and the individual's physiology may also affect HRV results and were not accounted for during this study. In spite of these restrictions, the study has successfully revealed the usefulness of ECG and HRV analysis in non-invasively evaluating ANS and monitoring the physiology of healthy college students.

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Appendix (Publication/Presentation)

A: Participant screening & selection(google form & questionnaire

The participant screening process was used to decide whether the participants were healthy college students appropriate for ECG and HRV analysis. The volunteers were first briefed about the purpose, methodology and physiological aspects of the study. A Perceived Stress Scale (PSS-10) and a general health questionnaire were used to screen participants. The questionnaire contained information about cardiovascular health, use of medications, chronic illness, and lifestyle factors that might affect HRV.

Study participants with cardiovascular disorders, chronic medical conditions, and medications which affect autonomic nervous system function were not included. Only those with low perceived stress scores (PSS $\leq$ 13) were used for the final analysis. Following screening and exclusion criteria, 10 healthy individuals were selected for recording their ECG under normal resting conditions and performing HRV analysis.

To ensure homogeneity in the study population and to reduce physiological variation due to pre-existing medical conditions or psychological disorders, the participant selection process was done.

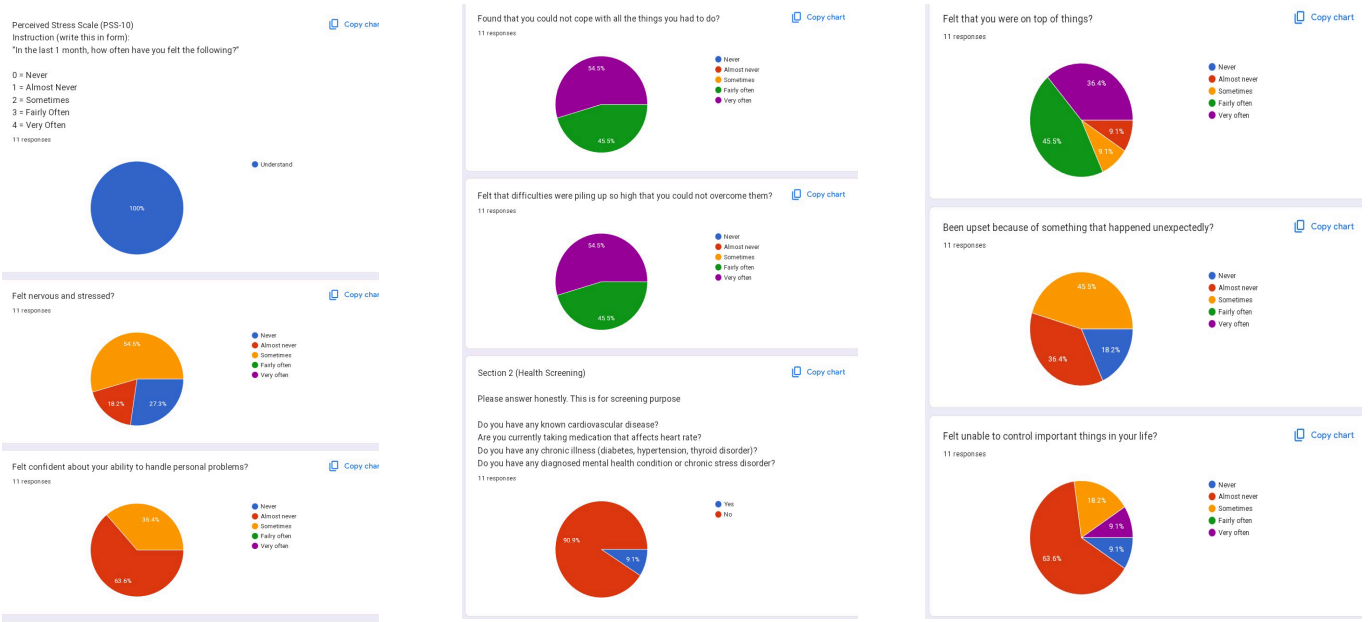
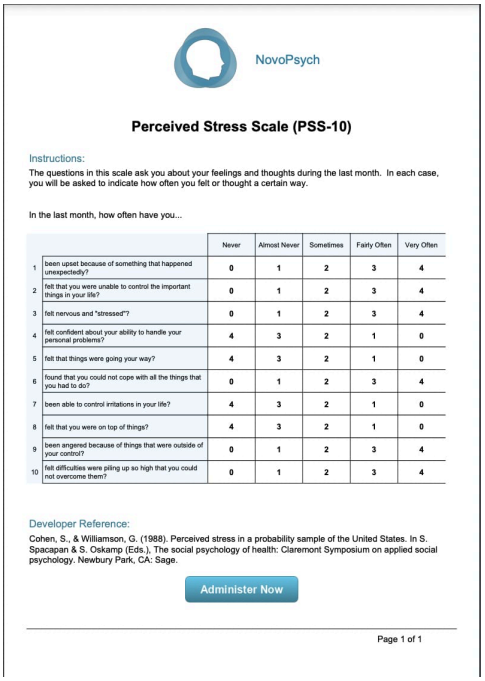


Figure A.1 : Google Form-Based Participant Screening Responses

1) Has your doctor ever said that you have a heart condition ☐ **OR** high blood pressure ☐?	☐	☐
2) Do you feel pain in your chest at rest, during your daily activities of living, **OR** when you do physical activity?	☐	☐
3) Do you lose balance because of dizziness **OR** have you lost consciousness in the last 12 months? Please answer **NO** if your dizziness was associated with over-breathing (including during vigorous exercise).	☐	☐
4) Have you ever been diagnosed with another chronic medical condition (other than heart disease or high blood pressure)? **PLEASE LIST CONDITION(S) HERE:** _____	☐	☐

Figure A.2 : self made Health Screening Questionnaire



NovoPsych

Perceived Stress Scale (PSS-10)

Instructions:

The questions in this scale ask you about your feelings and thoughts during the last month. In each case, you will be asked to indicate how often you felt or thought a certain way.

In the last month, how often have you...

		Never	Almost Never	Sometimes	Fairly Often	Very Often
1	been upset because of something that happened unexpectedly?	0	1	2	3	4
2	felt that you were unable to control the important things in your life?	0	1	2	3	4
3	felt nervous and "stressed"?	0	1	2	3	4
4	felt confident about your ability to handle your personal problems?	4	3	2	1	0
5	felt that things were going your way?	4	3	2	1	0
6	found that you could not cope with all the things that you had to do?	0	1	2	3	4
7	been able to control irritations in your life?	4	3	2	1	0
8	felt that you were on top of things?	4	3	2	1	0
9	been angered because of things that were outside of your control?	0	1	2	3	4
10	felt difficulties were piling up so high that you could not overcome them?	0	1	2	3	4

Developer Reference:

Cohen, S., & Williamson, G. (1988). Perceived stress in a probability sample of the United States. In S. Spacapan & S. Oskamp (Eds.), The social psychology of health: Claremont Symposium on applied social psychology. Newbury Park, CA: Sage.

Administer Now

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Figure A.3 : Perceived Stress Scale (PSS-10) Questionnaire

SCREENING SCORING TABLE

Participant	Score	Cardiovascular Diseases
P1	11	No
P2	6	No
P3	10	No
P4	12	No
P5	7	No
P6	9	No
P7	9	No
P8	5	No
P9	13	No
P10	8	No
P11	28	Yes

Table A.1 Participant Screening and Cardiovascular Health Status Table

B: ECG Data Acquisition Using BIOPAC System

The BIOPAC MP36 physiological data acquisition system with the ECG-1 channel configuration was used to record electrocardiograms (ECGs). It was decided that the BIOPAC system would be used for its reliability, ability to monitor physiological parameters in real time, and extensive use in biomedical and physiological research.

To ensure participants did not move un-necessarily during the data acquisition, they were asked to be calm beforehand. The electrodes were attached to the chest in a modified Lead II position, to capture the clear cardiac electrical activity. The right arm (RA) was used as the ground, the left arm (LA), left leg (LL) and right leg (RL) were used for electrodes.

In order to minimize the effects of motion artifacts and external disturbances, participants were recorded in a comfortable resting position in a quiet room. An ECG was recorded for about 25-30 seconds each. The BIOPAC software was used to convert analog physiological signals to digital for processing and storage.

The acquired ECG waveforms showed good R-peaks and the RR-intervals were successfully extracted and HRV analysis was performed in the Kubios HRV Scientific Lite 4.3.0 software.

C: HRV Analysis Using Kubios HRV Scientific Lite 4.3.0

The analysis of Heart Rate Variability (HRV) was performed with Kubios HRV software Scientific Lite version 4.3.0. ECG signals acquired from the BIOPAC system were exported and imported to the Kubios software for RR interval analysis and assessment of ANS.

Advanced signal processing tools such as artifact correction, RR interval extraction, time domain analysis, frequency domain analysis and nonlinear HRV analysis were accomplished using Kubios software. The software was able to automatically detect the R-peaks and create RR interval series for each participant.

The time domain parameters that were analyzed were the Mean Heart Rate (Mean HR), Minimum Heart Rate (Min HR), Maximum Heart Rate (Max HR), Standard Deviation of NN intervals (SDNN), and Root Mean Square of Successive Differences (RMSSD). Other autonomic measures, such as Parasympathetic Nervous System (PNS) Index and Sympathetic Nervous System (SNS) Index were also assessed.

To assess autonomic balance the frequency domain analysis tests for LF power, HF power and LF/HF ratio. Poincaré Plot generation, SD1 and SD2 measurements were used for nonlinear analysis to assess short-term heart rate variability (HRV) and long-term HRV patterns.

Cardiovascular autonomic regulation was visualized and interpreted using the software-generated graphs, histograms, RR interval plots and nonlinear plots.

D: HRV Parameters and Physiological Interpretation

The HRV parameters were evaluated using standard physiological ranges from the scientific literature, and as obtained from Kubios software. Within the range of moderate HRV, balanced sympathetic and parasympathetic activity, and cardiovascular regulation is associated with normal autonomic function.

Increased SDNN and RMSSD values are generally considered as evidence of autonomic adaptability and parasympathetic nervous system functioning, while decreases are seen as signs of stress, sympathetic overactivity or decreased CV flexibility.

The PNS index relates to parasympathetic activity and recovery status and the SNS index relates to activation of the sympathetic nervous system. Normal autonomic balance has moderate SNS values and normal PNS activity. The nonlinear Poincaré plots created by Kubios analysis showed relatively organized elliptical configurations in subjects indicating healthy cardiac autonomic function and a stable RR interval variability at rest.

The overall HRV results of the present study showed that there were no significant physiological stress for the subjects during recording of normal-day HRV.

E: Examination Day Recording Limitation and Future Scope

The original study design involved both routine rest and stress (examination day) conditions with ECG and HRV recordings for comparison for physiological analysis. The BIOPAC system, however, malfunctioned during the examination period and this prevented the experimental completion of examination-day ECG recordings.

Therefore, normal day ECG recordings and HRV parameters were only used in the present study. Physiological reactions to the tests were discussed as reported in previously published scientific literature.

Past studies show a relationship between examination stress and heart rate, an increase in sympathetic activity, a decrease in parasympathetic activity, a decrease in HRV, and an increase in LF/HF ratio. The physiological responses are regarded as markers of autonomic imbalance resulting from stress.

The future studies can be extended to direct acquisition of ECG during the exam day, with the use of functional physiological recording systems, thus allowing the acquisition of HRV data in real time, and in a stress-related context.

The comparison of a resting condition with examination stress condition may give a deeper understanding of autonomic nervous system regulation of college students when exposed to examination stress.

F: Further Graphical Analysis and Visualisation

The findings of physiology were interpreted and presented using multiple graphical outputs generated by the BIOPAC and Kubios HRV software. These included trend graphs of RR intervals, histogram of HRV, frequency domain spectra, nonlinear Poincaré plots, and ANS indices.

The graphical displays helped to better understand the patterns of cardiovascular variability and autonomic regulation in participants. ECG waveforms were used to determine the stability of cardiac rhythm in resting state and HRV plots were used to assess normal physiological variability.

The autonomic balance was suggested by frequency-domain graphs, with the LF and HF distributions. Measures from non-linear Poincaré plots showed organized elliptical shapes, suggesting the presence of stable RR interval variability and cardiac adaptability.

The visual outputs have made significant contributions to the physiological interpretation of the thesis and the quality of presentation of the thesis as a scientific study.

USER MANUAL

Step-by-Step ECG and HRV Analysis Procedure

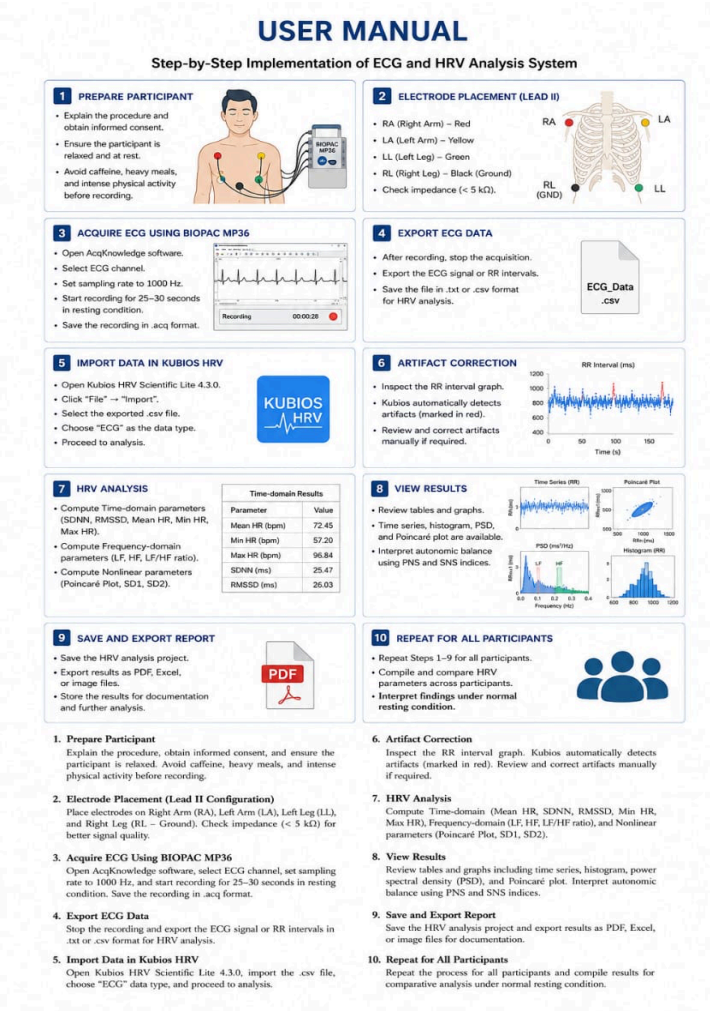


Figure M.1 : User Manual for ECG and HRV Analysis Procedure

Step 1: Setup BIOPAC System

Connect the BIOPAC MP36 system to the computer and open AcqKnowledge software. Select the ECG-1 channel and set the sampling rate to 1000 Hz.

Step 2: Prepare Participant

Explain the procedure and obtain informed consent. Ensure the participant is relaxed and seated comfortably.

Step 3: Attach Electrodes

Place electrodes using modified Lead II configuration:

RA – Right Arm

LA – Left Arm

LL – Left Leg

RL – Right Leg (Ground)

Step 4: Record ECG Signal

Start ECG recording in BIOPAC software. Record the ECG signal for 25–30 seconds under resting condition.

Step 5: Save ECG Data

Stop recording and save the ECG file in .acq format. Export RR interval or ECG data in .txt or .csv format.

Step 6: Open Kubios HRV Software

Launch Kubios HRV Scientific Lite 4.3.0 software and import the exported ECG/RR interval file.

Step 7: Perform Artifact Correction

Inspect RR interval signals and apply automatic artifact correction if necessary.

Step 8: Analyze HRV Parameters

Extract HRV parameters including:

Mean HR

Min HR

Max HR

SDNN

RMSSD

LF/HF Ratio

PNS Index

SNS Index

Step 9: Visualize Results

Observe RR interval plots, histograms, frequency-domain graphs, and Poincaré plots generated by Kubios software.

Step 10: Document Findings

Record all ECG and HRV results in tables and graphs for interpretation and thesis presentation.

Step 11: Interpret Physiological Condition

Interpret autonomic nervous system activity based on HRV parameters and compare findings with standard physiological ranges.

Step 12: Future Scope

Examination-day recordings could not be completed due to BIOPAC technical malfunction; therefore, examination-related analysis is discussed as future scope based on literature findings.